Irrationality of Cubic Threefolds after One Stabilization

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Abstract

We prove that $X \times \mathbf{P}^1$ is irrational for every smooth complex cubic threefold X. Working in the ordinary, non-enhanced Hodge-atom package of Katzarkov–Kontsevich–Pantev–Yu, we isolate the atom carried by the double zero packet of the cubic small quantum connection and introduce a rank-two atomic residue discriminant $\delta^\sharp$. A canonical elementary modification of the even rank-two block gives $\delta^\sharp = 4/9$ for the cubic atom. The only curve whose atom could carry the same Hodge representation has genus five, and there $\delta^\sharp = 0$; surface representatives are excluded by parity ranks and the classification of minimal surfaces. The ordinary Hodge-atom non-rationality criterion then obstructs rationality after one stabilization. Kuznetsov's birational correspondence extends the conclusion to every smooth prime Fano threefold of genus eight.

We also study the finer invariant ν_6, the number of primitive-sixth framed formal-monodromy eigenvalues of the numerical small even quantum connection. We compute $\nu_6(X) = 2$ and $\nu_6(X \times \mathbf{P}^1) = 4$, and establish product, blowup, and projective-bundle formulas, the last two under two stated hypotheses on the reconstruction tail and on divisor-tagging specialization. Under those hypotheses ν_6 is birationally invariant through dimension four, giving a second proof of the theorem.

Finally, an integral six-axis construction proves universal CH_0-triviality for every smooth member of the nonstandard A_5-invariant pencil: an all-degree saturation theorem for marked finite-étale graph quotients of elliptic powers makes the primitive minimal class of the intermediate Jacobian an integral divisor product. All but finitely many moduli points of this pencil lie outside the separated-variable locus covered by Colliot-Thélène's criterion. Universal CH_0-triviality and irrationality after one stabilization therefore coexist in an explicit nontrivial family, as they also do for Voisin's loci, the Fermat cubic, and the coprime-degree family of Yang–Yu–Zhu.

Keywords. cubic threefold; universal CH_0-triviality; irrationality; Hodge atoms; intermediate Jacobian; quantum connection; formal monodromy; weak factorization.

1 Introduction

A smooth projective variety is universally CH_0-trivial if the degree map

$$\deg : \mathrm{CH}_0(Y_L) \longrightarrow \mathbf{Z}$$

is an isomorphism for every field extension $L/\mathbf{C}$. This condition is a central consequence of stable rationality, and its failure is a standard stable-rationality obstruction. Its validity, however, need not provide a rational parametrization. We prove that even after one explicit stabilization it does not predict rationality for smooth cubic threefolds.

Theorem 1.1 (One-step irrationality)**.** *For every smooth complex cubic threefold X, the fourfold $X \times \mathbf{P}^1$ is irrational.*

The proof uses ordinary Hodge atoms in the sense of Katzarkov–Kontsevich–Pantev–Yu [22]. Their blowup and projective-bundle formulas package weak factorization into a free abelian ledger on atoms, and their ordinary non-rationality criterion says that a smooth d-fold containing an atom not realizable in dimension at most $d-2$ is irrational. We construct such an atom for a cubic threefold. At the small hyperplane point, the two-dimensional even zero packet of the Euler multiplication is a single Jordan block. A regular- F-manifold discriminant argument shows that the corresponding branch is indeed a connected component of the unramified spectral cover, rather than an accidental coalescence at one point. On its even rank-two atomic F-bundle, a canonical elementary modification turns the nilpotent leading block into a logarithmic one. The Poincaré pairing forces this modification to be regular and flatness forces its residue to move by conjugation. The resulting residue discriminant $\delta^\sharp$ is an ordinary atom invariant. For the cubic atom

$$\delta^\sharp = \frac{4}{9},$$

whereas the only possible curve representative, a genus-five curve, has $\delta^\sharp = 0$. Surface representatives are excluded by parity rank and the classification of minimal surfaces. Since $\mathsf{CF}(X \times \mathbf{P}^1) = 2\mathsf{CF}(X)$, the cubic atom occurs on the stabilized fourfold and the ordinary atom criterion proves Theorem 1.1.

The theorem is unconditional relative to the numbered ordinary Hodge-atom theorems imported from [22], in the usual sense that a proof is relative to its cited external results. At the time of writing that source is an unrefereed preprint, and its projective-bundle theorem ultimately uses the Iritani–Koto decomposition theorem [21]. No enhanced atom, Euler pairing, Serre automorphism, or non-archimedean Riemann–Hilbert statement is used in our proof.

The conclusion propagates immediately through stabilized birational correspondences. In particular, if $Y \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$ for a smooth cubic threefold X, then $Y \times \mathbf{P}^1$ is irrational. Kuznetsov's construction for a smooth prime Fano threefold V of genus eight gives rank-two projective bundles $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$ related by a flop [23, Theorems 2.17–2.18]; hence $V \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$ and is also irrational. The finer equality $\nu_6(V) = 2$ below remains conditional on the reconstruction-tail and divisor-tagging hypotheses of Section 5.

Unconditional separation results

Each separation statement rests on two independent inputs. Irrationality comes from the cubic Hodge atom. Universal CH_0-triviality is proved for the same cubic by a classical or cycle-theoretic theorem, and then passes to the product by the projective-bundle formula for Chow groups. Thus the vanishing of the usual decomposition-of-the-diagonal obstruction and the atom obstruction read genuinely different structures on the same fourfold.

Universally CH_0-trivial cubic threefolds are known in abundance. Voisin proves that the ten-dimensional moduli space of smooth cubic threefolds contains a nonempty countable union of subvarieties of codimension at most three parametrizing threefolds with universally trivial CH_0 [30, Theorem 4.5].

Corollary 1.2 (Separation in codimension three)**.** *There is a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds such that, for every member X, both X and $X \times \mathbf{P}^1$ are universally CH_0-trivial, while $X \times \mathbf{P}^1$ is irrational.*

Independently, Colliot-Thélène proves that a smooth cubic hypersurface whose equation is a sum of forms in separated groups of at most three variables is universally CH_0-trivial [11], which applies to the Fermat equation directly.

Corollary 1.3 (Fermat). *The fourfold*

$$\{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \times \mathbf{P}^1$$

is universally CH_0-trivial and irrational.

Yang–Yu–Zhu construct a two-dimensional family of smooth cubic threefolds admitting unirational parametrizations of degrees two and three [32, Theorems 1.1 and 3.3]. Parametrizations of coprime degrees give a decomposition of the diagonal, so every member of that family is universally CH_0-trivial; the same degrees persist on $X \times \mathbf{P}^m$ for every m [32, Corollary 3.5].

Corollary 1.4 (Coprime-degree cubics). *Let X be one of the cubic threefolds of [32, Theorem 3.3]. Then $X \times \mathbf{P}^1$ admits unirational parametrizations of coprime degrees, is universally CH_0-trivial, and is irrational.*

Yang–Yu–Zhu ask whether a variety with unirational parametrizations of coprime degrees can fail to be stably rational [32, Question 1.4]. Corollary 1.4 settles the first nontrivial stabilization for their examples. It does not settle stable irrationality: our atom obstruction ceases to exclude rationality from the second stabilization onward because a four- or five-dimensional stabilized variety may be allowed atoms represented in dimension three, and the cubic atom itself already has such a representative.

For our second separation result we give a different uniform proof of universal CH_0-triviality along the nonstandard A_5-invariant pencil. There the primitive minimal class of the intermediate Jacobian is algebraic by an integral saturation theorem for the six-axis lattice, for every member at once. This construction also reaches beyond the separated-variable criterion of Colliot-Thélène: all but finitely many moduli points represented by the pencil are not projectively equivalent to a cubic of separated-variable type.

Theorem 1.5 (Separation on the A_5-pencil). *Let W_5 be the five-dimensional irreducible A_5-module, and let B° be the smooth locus inside*

$$\mathbf{P}\big((\mathrm{Sym}^3 W_5^*)^{A_5}\big).$$

For the resulting family $\mathcal{X} \to B^\circ$ and every $b \in B^\circ(\mathbf{C})$,

$$\mathcal{X}_b \text{ and } \mathcal{X}_b \times \mathbf{P}^1 \text{ are universally } CH_0\text{-trivial},$$

and $\mathcal{X}_b \times \mathbf{P}^1$ is irrational. The family is non-isotrivial in coarse moduli. All but finitely many points of its coarse-moduli image are not represented by cubics whose defining form is a sum of cubic forms in disjoint groups of at most three variables.

The invariant cubic space in Theorem 1.5 is two-dimensional. Indeed, the character of W_5 is $(5, 1, -1, 0, 0)$ on the classes $1, 2A, 3A, 5A, 5B$, and the symmetric-cube character formula gives

$$\dim\big(\mathrm{Sym}^3 W_5^*\big)^{A_5} = 2.$$

Thus its projectivization is a pencil with connected smooth locus B°. Its coarse-moduli image is Hartlieb's M_{H_1}; this is not the component associated with $\mathbf{1} \oplus W_4$ [17, Lemma 5.5]. Since the family is non-isotrivial, its image in coarse moduli is a curve, one dimension below the locus of [32].

Remark 1.6. The pencil contains the Fermat cubic threefold; it is one of the two members that also admit the permutation action of A_5 [17, Lemma 5.5]. The mechanism is that W_5 is induced from a nontrivial cubic character χ of a point stabilizer $A_4 \subset A_5$: Frobenius reciprocity gives

$$\langle \mathrm{Ind}_{A_4}^{A_5} \chi,\ W_5 \rangle = \langle \chi,\ \mathrm{Res}_{A_4} W_5 \rangle = 1,$$

and both sides have dimension five, so $\mathrm{Ind}_{A_4}^{A_5} \chi \cong W_5$. Induction from a character realizes W_5 by monomial matrices whose nonzero entries are cube roots of unity, and such a matrix fixes $x_1^3 + \cdots + x_5^3$. Corollary 1.3 is therefore the member of Theorem 1.5 at which universal CH_0-triviality is also classical, and Yang–Yu–Zhu give that member an explicit degree-three unirational parametrization [32, Remark 3.6].

A finer conditional framed invariant

The ordinary atom proof above is deliberately minimal: it follows one rank-two cubic atom and uses only the finite residue discriminant needed to exclude low-dimensional representatives. We also study a finer invariant of the entire numerical small even quantum connection. Definition 5.1 counts the primitive-sixth eigenvalues of framed formal monodromy on the original loop disc and gives an invariant ν_6. The cubic packet calculation is unconditional and yields $\nu_6(X) = 2$. More generally, the product formula for Gromov–Witten theory gives $\nu_6(T \times \mathbf{P}^m) = (m+1)\nu_6(T)$, so in particular $\nu_6(X \times \mathbf{P}^1) = 4$ unconditionally.

The actual Iritani and Iritani–Koto comparison isomorphisms preserve the original-z-disc framing after coefficient extension. The string and divisor equations exactly remove the scalar degree-zero and fixed degree-two pieces of their reconstruction coordinates. We therefore split the remaining uncertainty into two statements in Section 5: Hypothesis 5.7R concerns only the residual reconstruction tail, while Hypothesis 5.7T is the separate specialization step in the divisor-tagging argument. The latter is already unnecessary for nef-canonical targets, for $\mathbf{P}^1$ and $\mathbf{P}^2$, for every rational geometrically ruled surface—for the Hirzebruch surface F_1, on specializations satisfying a mild condition that every blowup center satisfies—and, assuming 5.7R, for ruled surfaces over positive-genus curves; in the present low-dimensional proof it remains only for surface centers that are neither minimal nor geometrically ruled. A rank-two formal-germ theorem proves the local mechanism needed for the cubic packet, but we do not identify the reconstruction value with a point of that formal germ.

Conditional on Hypothesis 5.7R, ν_6 satisfies the displayed projective-bundle and blowup formulas. Conditional on Hypothesis 5.7R and on Hypothesis 5.7T for surface centers that are neither minimal nor geometrically ruled, it is birationally invariant through dimension four. That invariance reproves the one-step conclusion from the finer invariant.

Theorem 1.7 (One-step irrationality from the framed invariant)**.** *Assume Hypothesis 5.7R of Section 5, and Hypothesis 5.7T there only for surface centers that are neither minimal nor geometrically ruled. Let X be a smooth complex cubic threefold. Then $\nu_6(X \times \mathbf{P}^1) = 4$, every rational smooth projective fourfold has $\nu_6 = 0$, and consequently $X \times \mathbf{P}^1$ is irrational.*

Theorem 1.7 is not the logical basis of Theorem 1.1: it reaches the same conclusion under two hypotheses, through an invariant of the whole small even quantum connection rather than one atom. Its interest is that the invariant is finer. It separates a cubic threefold from projective three-space at the level of threefolds, where the atom argument gives nothing, and for genus eight it gives the conditional identity $\nu_6(V) = 2$.

Relation with earlier rationality results

Clemens and Griffiths proved that every smooth complex cubic threefold is irrational by means of its intermediate Jacobian $J(X)$ [8]. The direct Clemens–Griffiths mechanism no longer yields a contradiction after one stabilization. Indeed,

$$H^3(X \times \mathbf{P}^1, \mathbf{Q}) \simeq H^3(X, \mathbf{Q}), \qquad H^4(X \times \mathbf{P}^1, \mathbf{Q}) \simeq \mathbf{Q}(-2)^{\oplus 2}.$$

In a birational factorization of a fourfold, smooth surface centers may occur, and blowing up such a surface S contributes $H^1(S, \mathbf{Q})(-1)$ to H^3. For the cubic itself, the Fano surface of lines $F(X)$ satisfies $\mathrm{Alb}(F(X)) \simeq J(X)$ [8, Theorem 11.19], hence $H^1(F(X), \mathbf{Q})(-1) \simeq H^3(X, \mathbf{Q})$. Thus the surviving weight-three Hodge structure is already realizable as the H^1 of a smooth surface, exactly the form that a surface center may contribute, while the middle H^4 is Tate. The standard intermediate-Jacobian and blowup bookkeeping therefore gives no contradiction here, though this does not exclude more elaborate classical Hodge-theoretic obstructions. The ordinary Hodge-atom argument below refines this raw Hodge bookkeeping by following the quantum spectral packet carrying the Hodge representation rather than H^3 alone.

For comparison, Guéré obtains a projective-K3-type Hodge necessary condition for rational cubic fourfolds from evaluated quantum data [16], while Benedetti–Fay–Guéré–Manivel–Perrin prove a monodromy/coarse-atom irrationality criterion for certain Hodge-general Fano fourfolds with large vanishing middle H^4 [6]. Their hypotheses include $b_3 = 0$, so the criterion does not cover $X \times \mathbf{P}^1$, where $b_3 = 10$, the middle H^4 is Tate, and the surviving H^3 is already of surface-center type; the proof below instead requires finer information internal to the ordinary cubic atom.

The very general cubic threefold is now known to be stably irrational [12, Corollary 1.4], but stable rationality remains open for an arbitrary smooth cubic threefold. Cai recently recovered irrationality from the primitive-sixth-root formal monodromy of the cubic quantum connection [7]. Iritani and Iritani–Koto prove the quantum D-module decompositions for blowups and projective bundles [19, 21]; the ordinary Hodge-atom package of Katzarkov–Kontsevich–Pantev–Yu [22] packages these decompositions into birational atomic data. Theorem 1.1 uses that ordinary package together with the new rank-two residue discriminant, without the enhanced Serre-automorphism machinery of [22] and without transporting framed monodromy through reconstruction displacements.

On the cycle side, Voisin reduces universal CH_0-triviality of a cubic threefold to algebraicity of the primitive minimal class of its intermediate Jacobian [30]. Beckmann–de Gaay Fortman place such minimal classes in a general integral-Fourier framework [3]. For the very general cubic threefold the minimal class is not algebraic: every curve class on the intermediate Jacobian is an even multiple of it, so a very general cubic threefold admits no decomposition of the diagonal [12, Theorem 1.3 and Corollary 1.4]. Known pieces of the algebraic locus include Voisin's components of codimension at most three, the almost-diagonal cubics of [11], and the coprime-degree family of [32]. Theorem 3.5 places the nonstandard A_5-curve in the same algebraic-minimal-class locus by a uniform integral argument, and Proposition 3.9 shows that all but finitely many moduli points on that pencil lie outside the separated-variable locus covered by [11]. Thus for a general member the six-axis construction accounts for universal CH_0-triviality in a case not covered by that criterion. Proposition 3.8 places all but finitely many moduli points of that pencil outside the explicit family of [32, Theorem 3.3] as well: every member of their normal form carries an Eckardt point, and the general member of the pencil carries none. On Voisin's side, Proposition 2.5 shows that the intermediate Jacobian of the geometric generic member receives no odd-degree isogeny from a product of elliptic curves, although it is isogenous to the fifth power of an elliptic curve; her construction of components therefore does not reach this pencil through a product of factors of dimension at most three. We do not claim that the A_5-curve is disjoint from her locus. That would need both of the routes left open there to fail: the four-dimensional factor of Proposition 2.5 must not be the Jacobian of an irreducible curve of genus four, and the intermediate Jacobian must not be odd-degree isogenous to the Jacobian of an irreducible curve of genus five.

Proof map

Section 2 computes the six-axis polarization, identifies the exotic cubic gluing, and draws from it the restriction on product factorizations of the intermediate Jacobian. Section 3 proves the local divisor-product theorem, applies it to the primitive minimal class, and shows that the resulting A_5-curve is generically outside both the separated-variable locus and the explicit coprime-degree family. Section 4 proves Theorem 1.1 unconditionally from ordinary Hodge atoms and a rank-two residue discriminant, with the small cubic block reduction isolated for later reuse. Section 5 defines the finer invariant ν_6, proves exact low-degree shift and projective-space product formulas, records the formal-germ rank-two rigidity theorem and the two residual hypotheses, and computes $\nu_6(X) = 2$ and $\nu_6(X \times \mathbf{P}^1) = 4$ from the same cubic packet. The final section combines the unconditional irrationality theorem with the cycle-theoretic results and records the exact scope of the conditional framed refinement.

2 The six-axis polarization and the exotic packet

Let Ω be the six conjugate D_5-subgroups of A_5, equivalently the six Sylow-5 subgroups with their normalizers, and put

$$H_p = \mathrm{Aug}(\mathbf{F}_p^{\Omega})/\langle \mathbf{1} \rangle \qquad (p = 2, 3).$$

We record the modular representation calculation used throughout the six-axis argument, so that the commutants do not have to be inferred from a table.

Lemma 2.1 (The six-point hearts)**.** *For $p = 2, 3$, the $\mathbf{F}_p A_5$-module H_p is simple, and*

$$\mathrm{End}_{A_5}(H_2) = \mathbf{F}_4, \qquad \mathrm{End}_{A_5}(H_3) = \mathbf{F}_3.$$

Proof. Identify the six-point action with that of $A_5 \simeq \mathrm{PSL}_2(\mathbf{F}_5)$ on $\mathbf{P}^1(\mathbf{F}_5)$, whose points are $\infty, 0, 1, 2, 3, 4$; the stabilizer of a point is the normalizer of a Sylow-5 subgroup. Let $a(z) = z + 1$ and $s(z) = -1/z$, which generate $\mathrm{PSL}_2(\mathbf{F}_5)$. In $H_p = \mathrm{Aug}(\mathbf{F}_p^{\Omega})/\langle \mathbf{1} \rangle$, use the classes of

$$e_\infty - e_4, \quad e_0 - e_4, \quad e_1 - e_4, \quad e_2 - e_4$$

as a basis. The two generators act, for $p = 2$, by

$$A_2 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad S_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and, for $p = 3$, by

$$A_3 = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \qquad S_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

In both characteristics A_p has minimal polynomial $\Phi_5(t) = t^4 + t^3 + t^2 + t + 1$, which is irreducible over $\mathbf{F}_p$ because the order of p in $(\mathbf{Z}/5\mathbf{Z})^\times$ is four. Hence any nonzero A_5-stable subspace, being A_p-stable, is all of H_p; thus H_p is simple.

It remains to compute the commutant. Solving the linear equations $XA_p = A_pX$ and $XS_p = S_pX$ gives only scalar matrices for $p = 3$. For $p = 2$ it gives the two-dimensional algebra spanned by I and

$$T = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}, \qquad T^2 + T + I = 0.$$

Thus the two commutants are $\mathbf{F}_3$ and $\mathbf{F}_2[T]/(T^2 + T + 1) = \mathbf{F}_4$, respectively. This agrees with the modular matrix tables in [18]. $\square$

The six axes arise from the Fano surface. Let S be the Fano surface of a smooth member of the A_5-pencil and $(J, \Theta) = \mathrm{Alb}(S)$ its intermediate Jacobian. Each of the six D_5-subgroups gives an elliptic quotient $q_H : J \to E_H^{\mathrm{quot}}$ and its Rosati dual $i_H^{\mathrm{quot}} : E_H^{\mathrm{quot}} \to J$.
Generic non-CM axis. Hartlieb's elliptic isogeny factor, and hence every six-axis elliptic quotient, is non-CM at the geometric generic point of the pencil. Indeed, a CM endomorphism has a fixed imaginary-quadratic order, so the generic j-invariant is a root of its Hilbert class polynomial and is constant. After a finite base change the elliptic factor would then be isotrivial. Hartlieb's isogeny $J_b \sim E_b^5$ would put the whole intermediate-Jacobian family in the countable isogeny class of one fixed abelian variety. The image of an algebraic curve in a moduli space cannot be both positive-dimensional and countable: its image is constructible, and a positive-dimensional constructible subset contains a nonempty Zariski-open subset of an algebraic curve, hence has uncountably many complex points. This contradicts the fact that the closure of the intermediate-Jacobian image is a one-dimensional special subvariety [17, Proposition 5.7 and Remark 5.8]. Since the generic intermediate Jacobian is isotypic to E_b^5, each nonzero elliptic quotient is isogenous to E_b, and so is non-CM as well. This is a statement about the geometric generic point, not about every member: individual fibres may have complex multiplication, and there the divisor lattice selected by the marked presentation can be smaller than the full Neron–Severi lattice. Those fibres are covered by the prescribed-lattice reading of Theorem 3.3 recorded after its proof.

Proposition 2.2 (Generic six-axis polarization)**.** *On the geometric generic fibre of the nonstandard A_5-pencil, after coherent identification of the six quotient elliptic curves, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

Any five axes therefore give a generic-fibre isogeny $E^5 \to J$ whose pulled-back polarization has Smith type $(1, 6, 6, 6, 6)$.

Proof. The geometric generic elliptic factor has only integer endomorphisms. We recall the numerical passage from Roulleau's genus-two configuration to the six elliptic fibres. Let $D_g \subset S$ be the curve indexed by an involution $g \in A_5$, put $D_{\mathrm{tot}} = \sum_g D_g$, and, for a dihedral subgroup H of order ten, put

$$F_H = \sum_{g \in H,\, g^2 = 1} D_g.$$

Roulleau proves that F_H is a fibre of $S \to E_H^{\mathrm{quot}}$ [28, Theorem 11(D)], so $F_H^2 = 0$. His intersection table has $D_g^2 = -4$, and distinct curves meet with multiplicity zero, two, or one when the product of their labels has order two, three, or five, respectively; Lemma 14 proves that these values are constant in the corresponding families [28, Lemma 14, Corollary 16, and (3.1)]. Among the 105 unordered pairs of involutions, the product orders two, three, and five occur $15, 30, 60$ times. Hence

$$D_{\mathrm{tot}}^2 = 15(-4) + 2(30 \cdot 2 + 60 \cdot 1) = 180.$$

Every involution belongs to two of the six dihedral subgroups, and therefore $\sum_H F_H = 2D_{\mathrm{tot}}$. Two-transitivity on the six subgroups now gives

$$30(F_H \cdot F_{H'}) = (2D_{\mathrm{tot}})^2 = 720, \qquad F_H \cdot F_{H'} = 24 \quad (H \neq H').$$

By the preceding generic non-CM argument, the elliptic factor has only integer endomorphisms. After coherent identification of the generic elliptic quotients, two-transitivity makes the Rosati matrix have diagonal d and off-diagonal m. Put

$$D_H = q_H^*[0] \in \mathrm{NS}(J), \qquad N_H = i_H^{\mathrm{quot}} q_H \in \mathrm{End}(J).$$

Then $D_H|_S = F_H$, and N_H is the Rosati endomorphism attached to D_H. Coherent identification gives $q_H i_H^{\mathrm{quot}} = [d]$ and $q_H i_{H'}^{\mathrm{quot}} = [m]$ for $H \neq H'$. Each target identification is determined only up to a sign, and the ambiguity changes nothing used here. Replacing the chosen isomorphism $E_H^{\mathrm{quot}} \to E$ by its negative replaces q_H by $-q_H$ and, by functoriality of Rosati duality, i_H^{quot} by $-i_H^{\mathrm{quot}}$ at the same time. Hence $N_H = i_H^{\mathrm{quot}} q_H$ is unchanged, and so is $D_H = q_H^*[0]$, since $[-1]^*[0] = [0]$. The diagonal entry $q_H i_H^{\mathrm{quot}}$ is likewise unchanged, while the Gram matrix changes by a signed congruence DG_6D with $D = \mathrm{diag}(\pm 1)$, which alters neither the Smith type of the pulled-back polarization nor anything below. The coherent choice is the one making all off-diagonal entries equal; it is unique up to a global sign, which acts trivially. The invariant sum of the six axis maps vanishes because the five-dimensional constituent has no trivial summand; composing with q_H gives $d + 5m = 0$. The minimal-class identity $[S] = \Theta^3/3!$ [8], together with the polarized Riemann–Roch trace formula

$$\frac{\alpha\beta\Theta^{g-2}}{(g-2)!} = \mathrm{tr}(N_\alpha)\,\mathrm{tr}(N_\beta) - \mathrm{tr}(N_\alpha N_\beta),$$

obtained by polarizing the reduced-norm Riemann–Roch polynomial [15, Theorem 4.1], gives the following identities. Here tr is the degree-five trace in the coefficient representation, the trace entering that reduced norm, rather than the degree-ten trace on rational first homology:

$$\mathrm{tr}(N_H) = d, \qquad \mathrm{tr}(N_H N_{H'}) = m^2, \qquad F_H F_{H'} = d^2 - m^2.$$

Thus $d^2 - m^2 = 24$. Together with $d + 5m = 0$ and $d > 0$, this forces $(d, m) = (5, -1)$, hence $G_6 = 6I - J$. A five-by-five minor has Smith form $(1, 6, 6, 6, 6)$. □

Let B° be the connected smooth locus of the nonstandard A_5-pencil, and let $\mathcal{J} \to B^\circ$ be its relative intermediate Jacobian.

Lemma 2.3 (Relative six-axis source)**.** *There are an elliptic scheme $\mathcal{E} \to B^\circ$ and a relative isogeny*

$$f : \mathcal{A} := \mathcal{E} \otimes_{\mathbf{Z}} (\mathbf{Z}^\Omega / \mathbf{Z}\mathbf{1}) \longrightarrow \mathcal{J}$$

which is A_5-equivariant. The quotient lattice carries the form induced by $6I_6 - J_6$; both the source and this form carry their natural S_6-action. Moreover,

$$f^* \lambda_\Theta = \lambda_{\mathcal{A}}. \tag{2.1}$$

After omitting one axis, $\mathcal{A} \simeq \mathcal{E}^5$ and this form has matrix $6I_5 - J_5$. For $p = 2, 3$, put $\mathscr{V}_p = \mathcal{E}[p]$. The p-primary kernel of the source polarization has a natural A_5-equivariant symplectic description

$$\mathscr{D}_p \simeq H_p \otimes_{\mathbf{F}_p} \mathscr{V}_p \qquad (p = 2, 3). \tag{2.2}$$

The p-primary part $\mathscr{K}_p$ of $\ker f$ is an A_5-stable relative maximal isotropic subgroup of $\mathscr{D}_p$.

Proof. For $H \in \Omega$, the integral norm $n_H = \sum_{h \in H} h$ is a relative endomorphism of $\mathcal{J}$. Its action on rational first homology is ten times the projector onto the H-fixed summand. Fibrewise, Hartlieb's character calculation and isogeny decomposition give $H^1(J, \mathbf{Q}) \simeq W_5 \otimes H^1(E, \mathbf{Q})$, and hence the self-product of an elliptic curve through the five-dimensional module W_5 [17, Lemma 5.5 and Remark 5.8]. Since $\dim W_5^H = 1$, the fixed summand has rank two. More concretely, $n_H^2 = 10 n_H$. The scheme-theoretic image of a homomorphism of abelian schemes in characteristic zero is an abelian subscheme, and its quotient factorization commutes with base change [2, Theorem A]. That citation supplies the abelian-scheme structure; the relative dimension is pinned by $n_H^2 = 10 n_H$, which makes $n_H/10$ a projector, so on every fibre the image is the connected abelian subvariety it cuts out, of dimension one by $\dim W_5^H = 1$. Thus the scheme-theoretic norm image $\mathcal{E}_H^{\mathrm{im}} \subset \mathcal{J}$ is an elliptic scheme. Since $h n_H = n_H$ for every $h \in H$, the subgroup H acts trivially on $\mathcal{E}_H^{\mathrm{im}}$. Moreover $N_{A_5}(H) = H$, so transport by an element carrying H to H' is independent of the chosen transporter. The six norm-image elliptic schemes are therefore coherently isomorphic; fix one of them as $\mathcal{E}$.

Write $i_H^{\mathrm{im}} : \mathcal{E} \to \mathcal{J}$ for the transported primitive norm-image inclusions. On the geometric generic fibre, Hartlieb's isogeny decomposition identifies $\mathrm{Hom}^0(E, J)$ with the irreducible module W_5. The coherently transported maps $(i_H^{\mathrm{im}})_{H \in \Omega}$ form one A_5-orbit, so their sum is invariant. Since $W_5^{A_5} = 0$, that sum is zero. The connected smooth curve B° is integral, $\mathcal{E}$ is integral, and $\mathcal{J}$ is separated over B°. Two morphisms from an integral scheme to a separated one agree along a closed subscheme; here that subscheme contains the dense generic fibre, so $\sum_H i_H^{\mathrm{im}}$ and the zero map agree everywhere and $\sum_H i_H^{\mathrm{im}} = 0$ over B°. Hence $(i_H^{\mathrm{im}})_{H \in \Omega}$ descends from $\mathcal{E}^\Omega$ to the displayed map $f : \mathcal{A} \to \mathcal{J}$, and the construction is A_5-equivariant.

Roulleau constructs the D_5-fibration from an Albanese quotient $q_H : J \to E_H^{\mathrm{quot}}$ by a connected four-dimensional abelian subvariety [28, the sentence following Lemma 17 and the proof of Theorem 11(D)]. Thus $F_H = q_H^*[0]|_S$. The connected kernel makes the Rosati dual of q_H a primitive inclusion, and its image is the unique H-fixed elliptic subvariety. It therefore agrees, up to an automorphism of the target elliptic curve, with the norm-image elliptic curve $\mathcal{E}_H^{\mathrm{im}} \hookrightarrow \mathcal{J}$. This also identifies the norm-image axis, up to isogeny, with Hartlieb's generic elliptic isogeny factor. The quotient curve E_H^{quot} and the norm-image curve $\mathcal{E}_H^{\mathrm{im}}$ are therefore kept distinct until this point; the Rosati dual identifies the latter with the image of the former, up to the target automorphism just described. Consequently the generic-fibre classes computed in Proposition 2.2 are the classes of these relative homomorphisms. They give diagonal entry 5 and off-diagonal entry -1 on the geometric generic fibre. Applying the same generic-density and separatedness argument to these homomorphisms gives the identities over B°, hence (2.1). Choosing any five coordinate classes identifies the quotient lattice with $\mathbf{Z}^5$, where its matrix is $6I_5 - J_5$. Since this has Smith type $(1, 6, 6, 6, 6)$, the source polarization is nondegenerate, so f is a finite flat isogeny. From $\lambda_{\mathcal{A}} = f^\vee \lambda_\Theta f$ one gets

$$|\ker \lambda_{\mathcal{A}}| = 6^8, \qquad |\ker f| = \deg f = 6^4.$$

For $x, y \in \ker f$, functoriality of the polarization commutator pairing gives $e_{\lambda_{\mathcal{A}}}(x, y) =$

$e_{\lambda_\Theta}(fx, fy) = 1$. Thus $\ker f$ is isotropic and has square-root order in $\ker \lambda_{\mathcal{A}}$, hence is maximal isotropic. Its p-primary part has fibrewise order p^4 for $p = 2, 3$.

For $p = 2, 3$, the primary discriminant local system is

$$\mathscr{D}_p = \ker(6I_6 - J_6 : \mathcal{A}[p] \longrightarrow \mathcal{A}^\vee[p]).$$

Modulo p, the radical quotient of the induced coefficient form is canonically

$$\mathrm{Aug}(\mathbf{F}_p^\Omega)/\langle \mathbf{1} \rangle = H_p.$$

The form on this coefficient discriminant is the normalized form

$$b_p(\bar{x}, \bar{y}) = \frac{1}{p} x^{\mathrm{t}}(6I_6 - J_6)y \pmod{p} = \begin{cases} \bar{x} \cdot \bar{y}, & p = 2, \\ -\bar{x} \cdot \bar{y}, & p = 3. \end{cases} \tag{2.3}$$

For augmentation lifts, the product of the coordinate sums is divisible by p^2, which proves the displayed formulas. They are unchanged by adding the constant vector or p times an integral vector. The dot product on the augmentation module has radical exactly the constant line, so b_p is nondegenerate on H_p; b_2 is alternating and b_3 is symmetric. Tensoring with the Weil pairing on $\mathscr{V}_p$ gives (2.2) as a symplectic isomorphism. Since f is A_5-equivariant, each $\mathscr{K}_p$ is A_5-stable. The preceding order and isotropy calculation makes it a relative maximal isotropic subgroup of this rank-eight local system. □

Consequently the generic Gram calculation of Proposition 2.2 extends, by the relative homomorphisms and separatedness in Lemma 2.3, to every smooth fibre of the pencil. Thus every smooth fibre has six-axis Rosati matrix $6I_6 - J_6$, and any five axes have Smith type $(1, 6, 6, 6, 6)$.

Proposition 2.4 (Principal gluing packet)**.** *The A_5-stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

The three $\mathbf{F}_2$-rational kernels are S_6-stable. Each of the other two has stabilizer exactly A_5, and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard A_5-cubic component uses the latter pair. The A_5-stable three-primary kernels form $\mathbf{P}^1(\mathbf{F}_3)$; after choosing a complementary symplectic ruling, each is a scalar graph.

Proof. At two, by Lemma 2.1, the coefficient discriminant is the simple four-dimensional $\mathbf{F}_2 A_5$-module H_2, whose commutant is $\mathbf{F}_4$. The full discriminant is the semisimple module $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2 \simeq H_2^{\oplus 2}$. Every stable half is therefore one copy of H_2, and Schur's lemma identifies such submodules with the one-dimensional $\mathbf{F}_4$-subspaces of $\mathbf{F}_4^2$. This gives $\mathbf{P}^1(\mathbf{F}_4)$.

Here is the polarization check explicitly. Identify the coefficient heart with the natural two-dimensional $\mathbf{F}_4$-module. The trace-determinant form is a nonzero invariant alternating form. Relative to it, the normalized form b_2 of (2.3) differs by a nonzero scalar in $\mathbf{F}_4$; rescaling an $\mathbf{F}_4$-basis absorbs that scalar because squaring permutes $\mathbf{F}_4^\times$. We may therefore put

$$b(x, y) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2} \det(x, y).$$

On two multiplicity copies use

$$\Omega((x, y), (x', y')) = b(x, y') + b(y, x').$$

Every scalar $a \in \mathbf{F}_4$ is self-adjoint for b, since $b(ax, y) = b(x, ay)$. Thus the vertical line and all four graphs Γ_a are isotropic of half dimension. The five module-theoretic possibilities are exactly the five polarized gluings; no further quadratic condition removes the exotic pair.

The simplex group acts on both multiplicity copies of the coefficient heart over $\mathbf{F}_2$. Hence the three lines defined over $\mathbf{F}_2$ retain S_6. The stabilizer of an exotic graph is equally intrinsic. A permutation that fixes Γ_ω commutes with ω, hence is $\mathbf{F}_4$-linear on the two-dimensional $\mathbf{F}_4$-space H_2. If its determinant is $d \in \mathbf{F}_4^\times$, preservation of b says

$$\mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(dz) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(z) \qquad (z \in \mathbf{F}_4).$$

The trace pairing is nondegenerate, so $d = 1$. It therefore lies in $\mathrm{SL}_2(4)$, of order 60. The given A_5 already has that order and lies in the stabilizer, proving equality. The odd coset in its order-120 normalizer is semilinear and applies field Frobenius, so it exchanges ω and ω^2.

To pass from the generic fibre to the family, put

$$\mathscr{P} = \mathbf{P}_{\mathbf{F}_4}(\mathbf{F}_4 \otimes_{\mathbf{F}_2} \mathscr{V}_2).$$

By Lemma 2.3, this is the five-sheet local system of A_5-stable principal halves. It has the degree-three sub-local system $\mathscr{P}_{\mathrm{rat}} = \mathbf{P}_{\mathbf{F}_2}(\mathscr{V}_2)$ and the degree-two complement $\mathscr{P}_{\mathrm{ex}}$. Here H_2, together with its commutant $\mathbf{F}_4$, is constant, while monodromy acts only on $\mathscr{V}_2$. Its action is therefore $\mathbf{F}_4$-linear, not semilinear, and factors through $\mathrm{GL}_2(\mathbf{F}_2)$; in particular it preserves both subsets. The relative kernel $\mathscr{K}_2$ gives a section of $\mathscr{P}$.

At the geometric generic point this section cannot be rational. Otherwise the natural faithful S_6-action on $\mathcal{A} = \mathcal{E} \otimes (\mathbf{Z}^\Omega/\mathbf{Z}\mathbf{1})$, which is the identity on $\mathcal{E}$ tensored with the faithful permutation action on the quotient lattice, would preserve the rational kernel: under (2.2) it acts on H_2 and trivially on $\mathscr{V}_2$, so every member of $\mathscr{P}_{\mathrm{rat}}$ is S_6-stable. At three, Lemma 2.1 says that the analogous coefficient heart H_3 is simple and $\mathrm{End}_{A_5}(H_3) = \mathbf{F}_3$. The semisimple module $H_3^{\oplus 2}$ has stable halves parametrized by $\mathbf{P}^1(\mathbf{F}_3)$. The tensor pairing is b_3 times the alternating multiplicity pairing, so the three affine scalar graphs and the vertical line are all maximal isotropic. The natural S_6-action is the same on both multiplicity copies, so it fixes every one of these four lines. The dependency is thus: a rational two-primary half is S_6-fixed, every three-primary half is S_6-fixed, and $|\ker f| = 6^4$ makes the two primary parts exhaust $\ker f$. The full kernel would therefore be S_6-stable, so each $s \in S_6$ would descend to $\bar{s} : J \to J$. The descended map preserves the principal polarization. Indeed s preserves $6I_6 - J_6$, hence $s^*\lambda_{\mathcal{A}} = \lambda_{\mathcal{A}}$, and

$$f^*(\bar{s}^*\lambda_\Theta) = s^* f^* \lambda_\Theta = s^*\lambda_{\mathcal{A}} = \lambda_{\mathcal{A}} = f^*\lambda_\Theta,$$

while $\varphi \mapsto f^\vee \varphi f$ is injective on $\mathrm{Hom}(J, J^\vee)$ because f is an isogeny; so $\bar{s}^*\lambda_\Theta = \lambda_\Theta$ and $\bar{s} \in \mathrm{Aut}(J, \Theta)$. Everything here is a statement about polarization classes, so the translation and Appell–Humbert semicharacter choices that realize a class by an actual line bundle do not enter. The action would be faithful: if a source automorphism acts trivially on the quotient, the image of its difference from the identity is both connected and contained in the finite isogeny kernel, hence zero. Strong Torelli gives an exact sequence [17, Theorem 3.1]

$$1 \longrightarrow \{\pm 1\} \longrightarrow \mathrm{Aut}(J, \Theta) \longrightarrow \mathrm{Aut}(X) \longrightarrow 1.$$

The kernel of the resulting map from S_6 to $\mathrm{Aut}(X)$ would be a normal subgroup of order at most two; it is trivial. Thus S_6 would act faithfully on a smooth cubic threefold. This contradicts the classification of faithful automorphism groups [17, Theorem 2.1]. Five of the six listed ambient groups have order smaller than $|S_6| = 720$. The remaining group is $(\mathbf{Z}/3\mathbf{Z})^4 \rtimes S_5$, of order 9720, which is not divisible by 720. None can contain S_6. The section is therefore generically in $\mathscr{P}_{\mathrm{ex}}$. Its inverse image is open and closed in the connected base B°, so every smooth fibre has exotic two-primary kernel. The same calculation at three gives the four lines parametrized by $\mathbf{P}^1(\mathbf{F}_3)$; choosing any complementary line as the vertical ruling makes the selected half an affine scalar graph. □

For the cycle-theoretic argument, the unordered pair is enough. Both members act on the heart with the same squarefree minimal polynomial t^2+t+1. Theorem 3.5 uses only the conclusion that the geometric kernel is not one of the three $\mathbf{F}_2$-rational gluings. At three it uses only scalarity of the possible A_5-stable graphs, not the finer monodromy selection.

Hartlieb proves that the closure of its intermediate-Jacobian image is a one-dimensional special subvariety [17, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

The exotic kernel also limits how J can be assembled from lower-dimensional principally polarized pieces. Voisin's components of codimension at most three are produced by an odd-degree isogeny from the Jacobian of a possibly reducible curve [30, proof of Theorem 4.5]. A principally polarized abelian variety of dimension at most three is such a Jacobian, so the route is automatic whenever the intermediate Jacobian receives an odd-degree isogeny from a product of factors of dimension at most three under which the polarization pulls back to an odd multiple of the product polarization. Here it is not available.

Proposition 2.5 (No elliptic-product route)**.** *Let J be the intermediate Jacobian of the geometric generic fibre of the nonstandard A_5-pencil, let E be the geometric generic fibre of $\mathcal{E}$, and let*

$$\mu : \prod_{i=1}^{k}(A_i,\Theta_i) \longrightarrow (J,\Theta)$$

be an isogeny from a product of principally polarized abelian varieties with $\mu^*\Theta = m\sum_i \Theta_i$; its degree is m^5, since $\deg\lambda_{\mu^*\Theta} = (\deg\mu)^2\deg\lambda_\Theta$ while $\deg\lambda_{m\sum_i\Theta_i} = m^{10}$, so the degree is odd exactly when m is. If m is odd then either $k=1$, or $k=2$ and the two factors have dimensions one and four. The second case occurs: every axis $H\in\Omega$ gives such a μ, with A_1 an elliptic curve isogenous to E. Moreover J receives no odd-degree isogeny at all from a product of five elliptic curves, whatever polarizations those factors carry, although J is isogenous to E^5.

Proof. Write $\Lambda = \mathbf{Z}^\Omega/\mathbf{Z}\mathbf{1}$ with the symmetric form κ of matrix $6I_6 - J_6$ from Proposition 2.2, put $M = H_1(E,\mathbf{Z})$ and $L = H_1(J,\mathbf{Z})$, and let $\Lambda_1 = \Lambda^\vee \cap \frac{1}{2}\Lambda$, where $\Lambda^\vee$ is the κ-dual. Doubling identifies Λ_1/Λ with the kernel of κ modulo two inside $\Lambda/2\Lambda$, equivariantly, hence with the two-primary coefficient heart H_2 of (2.2); the $\mathbf{F}_4$-scalars and the pairing b are transported along that identification. By Lemma 2.3, f_* identifies $H_1(\mathcal{A},\mathbf{Z})$ on the geometric generic fibre with $\Lambda\otimes M \subseteq L$, with quotient $\ker f$, and the polarization form is the tensor product of κ with the symplectic form of M. Fix a basis of $M\otimes\mathbf{Z}_2$. By Proposition 2.4 the two-primary kernel is one of the two exotic graphs, so in that basis

$$L\otimes\mathbf{Z}_2 = \{(u,u')\in(\Lambda_1\otimes\mathbf{Z}_2)^{\oplus 2} : [u'] = \omega^{\pm 1}[u]\},$$

where $[\,\cdot\,]$ is the class in H_2 and ω is the $\mathbf{F}_4$-scalar of order three.

At the geometric generic point E has no complex multiplication, so the endomorphism algebra of the weight-one Hodge structure $M_{\mathbf{Q}}$ is $\mathbf{Q}$. As $H_1(J,\mathbf{Q}) = W_5\otimes M_{\mathbf{Q}}$, every sub-Hodge structure is $U\otimes M_{\mathbf{Q}}$ for a subspace $U\subseteq W_5$, and two of them are orthogonal exactly when the subspaces are κ-orthogonal.

Let $U_i \subseteq W_5$ be the subspace with

$$\mu_* H_1(A_i, \mathbf{Q}) = U_i \otimes M_{\mathbf{Q}},$$

and put $L' = \mu_*(\bigoplus_i H_1(A_i, \mathbf{Z}))$. It is an orthogonal direct sum of sublattices of the sub-Hodge structures $L \cap (U_i \otimes M_{\mathbf{Q}})$, each summand carrying m times a unimodular alternating form. The index of L' in L is odd because the degree of μ is, so localizing at two turns those inclusions into equalities: $L \otimes \mathbf{Z}_2$ is the orthogonal direct sum of the $P_i = (L \cap (U_i \otimes M_{\mathbf{Q}})) \otimes \mathbf{Z}_2$, each unimodular. Apply the displayed description of $L \otimes \mathbf{Z}_2$ in two ways. Its first-coordinate projection is all of $\Lambda_1 \otimes \mathbf{Z}_2$ and carries P_i into $(\Lambda_1 \cap U_i) \otimes \mathbf{Z}_2$; summing over i and comparing with that surjectivity gives $\Lambda_1 \otimes \mathbf{Z}_2 = \bigoplus_i (\Lambda_1 \cap U_i) \otimes \mathbf{Z}_2$, with equality in each piece. Its intersection with the first coordinate subspace is $\Lambda \otimes \mathbf{Z}_2$, and that subspace is the direct sum of its intersections with the $U_i \otimes M_{\mathbf{Q}}$, so $\Lambda \otimes \mathbf{Z}_2 = \bigoplus_i (\Lambda \cap U_i) \otimes \mathbf{Z}_2$. Hence $H_2 = \bigoplus_i H_2^{(i)}$ with $H_2^{(i)} = (\Lambda_1 \cap U_i)/(\Lambda \cap U_i)$. For $(u, u') \in P_i$ both classes lie in $H_2^{(i)}$ and, by that same projection, u runs over all of $\Lambda_1 \cap U_i$; so $\omega H_2^{(i)} \subseteq H_2^{(i)}$ and every summand is an $\mathbf{F}_4$-subspace.

That decomposition is orthogonal for the trace-determinant pairing b used above, and H_2 has $\mathbf{F}_4$-dimension two, so at most two summands are nonzero, with $\mathbf{F}_4$-dimensions 2 and 0, or 1 and 1. The second pattern is impossible. An $\mathbf{F}_4$-line $\ell = \mathbf{F}_4 x$ is totally isotropic for b, since $\det(x, ax) = a \det(x, x) = 0$, and b is nondegenerate, so $\ell^\perp = \ell$; two $\mathbf{F}_4$-lines orthogonal to one another therefore coincide, and cannot be complementary. One summand consequently carries the whole heart. The discriminant group of a lattice of rank r needs at most r generators and H_2 needs four, so that summand has $\dim U_i \geq 4$. This leaves $k = 1$, or $k = 2$ with dimensions four and one.

For the second case let $v \in \Lambda$ be the class of an axis, so $\kappa(v, v) = 5$. As 5 is a unit at two, $\Lambda \otimes \mathbf{Z}_2$ is the orthogonal sum of $\mathbf{Z}_2 v$ and $(\Lambda \cap v^\perp) \otimes \mathbf{Z}_2$, and the first summand is unimodular, so the whole coefficient heart is carried by the second. By the same display, $L \otimes \mathbf{Z}_2$ is then the orthogonal sum of the localizations of $L \cap (\mathbf{Q}v \otimes M_{\mathbf{Q}})$ and $L \cap (v^\perp \otimes M_{\mathbf{Q}})$, and both are unimodular, so the index of that sum in L is odd. On each summand the alternating form has odd elementary divisors $d_1 \mid \cdots \mid d_g$; passing to the sublattice spanned by $d_g e_i / d_i$ and f_i in a symplectic basis makes it d_g times a unimodular form, at the odd index $\prod_i (d_g/d_i)$. Taking m to be the product of the two resulting multipliers and shrinking each summand again by the complementary odd factor makes both forms m times a unimodular one. The first summand has rank two and gives an elliptic curve isogenous to E; the second has rank eight.

For the last assertion drop the polarization hypothesis and let $\mu : \prod_{i=1}^5 E_i \to J$ be any isogeny of odd degree from a product of elliptic curves. The argument above through $H_2 = \bigoplus_i H_2^{(i)}$ with each summand ω-stable uses only that the index is odd, not orthogonality, so it applies. Each U_i is now a line, so $H_2^{(i)} = (\Lambda_1 \cap U_i)/(\Lambda \cap U_i)$ is a quotient of a rank-one lattice pair and therefore cyclic; being killed by two and $\mathbf{F}_4$-stable, it has even dimension over $\mathbf{F}_2$, hence vanishes. Then $H_2 = 0$, against $|H_2| = 16$. □

Hartlieb records, in the footnote to [17, Remark 5.8], that van Geemen and Yamauchi split the intermediate Jacobian of a cubic threefold with an automorphism of order five, up to isogeny, as an elliptic curve times the square of an abelian surface. The proposition is compatible with that: their splitting is an isogeny, and what fails here is its realization by an odd-degree isogeny matching the polarizations. Two routes into Voisin's construction therefore remain, one for each case of the proposition: the four-dimensional factor of the second case may be the Jacobian of an irreducible curve of genus four, or J itself may be odd-degree isogenous to the Jacobian of an irreducible curve of genus five. Both are open, so this pencil is not known to lie in one of those components, and Section 3 does not depend on the question: the algebraicity of the primitive minimal class is proved there directly.

3 Integral divisor products and the primitive minimal class

Let (A, Θ) be a principally polarized abelian variety of dimension g. Put

$$P_A^k = \operatorname{im}\Big(\operatorname{Sym}^k \operatorname{NS}(A) \longrightarrow H^{2k}(A, \mathbf{Z})\Big), \qquad \gamma_A = \frac{\Theta^{g-1}}{(g-1)!}.$$

The class $D^k/k!$ of a divisor on an abelian variety is integral. The question here is whether these divided powers already belong to the smaller lattice P_A^k generated by ordinary products of integral divisors.

For quotients of elliptic powers, this question is controlled by the valuation pattern of the descended coefficient forms. The main local result of this section says that finite-étale graph slopes force all divided powers of the prescribed graph divisor lattice into ordinary products, in every degree. The proof has two steps: the graph equations produce a symmetric matrix-of-ideals lattice, and a midpoint inequality makes that lattice rank-one generated. Rank-one coefficient divisors have square zero, so their divided powers expand without factorials. The six-axis application supplies the primitive minimal class needed below.

3.1 A local graph theorem

Work over a complete unramified discrete valuation ring O, with uniformizer p. Fix an elliptic curve E, an isogeny $q : E^n \to Y$ to a principally polarized abelian variety, and extend the p-adic coefficient lattices to O. A symmetric matrix $A \in \operatorname{Sym}_n(O)$ denotes the standard coefficient divisor D_A on E^n, generated by coordinate and Poincare divisors. Define the prescribed graph-divisor lattice and its ordinary product image by

$$\begin{aligned} N_{\mathrm{gr}}(q, O) &= \{D \in \operatorname{NS}(Y) \otimes O : q^*D = D_A \text{ for some } A \in \operatorname{Sym}_n(O)\}, \\ P_{\mathrm{gr}}^k &= \operatorname{im}\Big(\operatorname{Sym}^k N_{\mathrm{gr}}(q, O) \longrightarrow H^{2k}(Y, \mathbf{Z}_p) \otimes O\Big). \end{aligned} \tag{3.1}$$

Because q^* is injective on torsion-free cohomology, we identify $N_{\mathrm{gr}}(q, O)$ with the symmetric matrices whose coefficient divisors descend through q. This is the prescribed lattice even when E has complex multiplication; no extra CM divisor is included.

Suppose the pullback coefficient polarization on the elliptic-power source is

$$G = PB, \qquad P = \bigoplus_a p^a I_{M_a}, \qquad B = \mathop{\perp}_a B_a, \tag{3.2}$$

where each B_a is unimodular. A *marked finite-étale graph presentation* consists of a symplectic ruling of the primary polarization kernel in which $\ker q$ is the graph of an endomorphism T, with the following properties: T preserves the depth summands M_a,

$$T^{\mathsf{t}} B^{-1} = B^{-1} T \quad \text{or, equivalently,} \quad BT^{\mathsf{t}} = TB, \tag{3.3}$$

and the algebra generated by T modulo p^a is finite etale on every positive-depth summand. Thus adjointness is taken for the dual coefficient form B^{-1}. No etaleness is required on M_0.

After a finite unramified extension splitting these etale algebras, the positive-depth summands decompose orthogonally into blocks

$$L_i = L_{a_i, \lambda_i}, \qquad T_i = t_i I + p^{a_i} S_i, \tag{3.4}$$

with S_i integral. Indeed, completeness lifts the primitive idempotents of each split finite-étale quotient to idempotents in the algebra generated by T. They are polynomials in T, so (3.3) makes them self-adjoint for B^{-1}. Their images are therefore mutually orthogonal for B^{-1}, and hence for B; on each image, T is scalar modulo the corresponding power p^{a_i}. We keep the unit part as one block. Write $\delta_{ij} = v_p(t_j - t_i)$, with $v_p(0) = \infty$, for two positive-depth blocks.

Lemma 3.1 (The graph coefficient lattice)**.** *In the split basis, the coefficient lattice of descended graph divisors is the symmetric matrix-of-ideals lattice whose diagonal block i has depth p^{a_i} and whose (i,j)-block has depth*

$$e_{ij} = \max\{a_i, a_j, a_i + a_j - \delta_{ij}\}. \tag{3.5}$$

For the unit block and a positive-depth block j, one has $e_{0j} = a_j$. In particular,

$$e_{ij} \geq \max(a_i, a_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil. \tag{3.6}$$

Proof. Choose the symplectic basis determined by the marked ruling and the depth decomposition. Adjoining the graph generators to the source homology lattice gives the quotient lattice, and in these coordinates the change-of-lattice matrix is

$$C = \begin{pmatrix} P^{-1} & P^{-1}T \\ 0 & I \end{pmatrix}.$$

A coefficient alternating form descends precisely when its matrix transported by C is integral. For a symmetric source coefficient A, direct multiplication gives

$$C \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} C^{\mathsf{t}} = \begin{pmatrix} P^{-1}(AT^{\mathsf{t}} - TA)P^{-1} & P^{-1}A \\ -AP^{-1} & 0 \end{pmatrix}. \tag{3.7}$$

Thus descent is equivalent to the integrality of the three displayed nonzero blocks. Here $A = A^{\mathsf{t}}$ is a symmetric coefficient bilinear form in this normalization. On two positive-depth blocks $L_i \times L_j$, the two off-diagonal blocks first force A_{ij} to have depth $\max(a_i, a_j)$. Substituting $T_i = t_i I + p^{a_i} S_i$ into the upper-left block leaves the additional condition

$$p^{-a_i - a_j}(t_j - t_i)A_{ij} \in M_{r_i \times r_j}(O);$$

the terms containing S_i, S_j are already integral. Intersecting the two conditions gives (3.5). Changing a scalar lift cannot change its maximum. Pairing the unsplit unit block with L_j gives $e_{0j} = a_j$ directly from (3.7), and (3.6) follows. □

The inequality (3.6) is the valuation shadow of the tropical positive semidefinite cone studied by Yu [31]. We need the following signed integral lift of that inequality; in particular, no division by two is permitted.

Lemma 3.2 (Rank-one generation over a DVR)**.** *Let L be a symmetric matrix-of-ideals lattice over O, with diagonal depths p^{a_i} and cross depths $p^{e_{ij}}$. Then L is generated by its rank-one forms if and only if*

$$2e_{ij} \geq a_i + a_j \qquad (i \neq j). \tag{3.8}$$

Proof. If $cvv^{\mathsf{t}} \in L$, the two diagonal conditions imply

$$v_p(cv_iv_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil.$$

Hence rank-one forms cannot generate a shallower cross ideal.

Conversely, fix a generator $p^e(uv^{\mathsf{t}} + vu^{\mathsf{t}})$ of a cross block. Choose $x \geq a_i$ with $y = 2e - x \geq a_j$. The integers x, y have the same parity. If $t = \min(x, y)$, write $x = t + 2r$ and $y = t + 2s$. Then

$$p^t(p^r u + p^s v)(p^r u + p^s v)^{\mathsf{t}} - p^x uu^{\mathsf{t}} - p^y vv^{\mathsf{t}} = p^e(uv^{\mathsf{t}} + vu^{\mathsf{t}}). \tag{3.9}$$

All three rank-one terms lie in L. Together with the diagonal rank-one forms, these identities generate L, without division by two. □

Theorem 3.3 (All-degree graph saturation)**.** *For a marked finite-étale graph presentation, every divided power of a class in its prescribed graph divisor lattice is an ordinary integral divisor product. Equivalently, after localization at p, the divided-power envelope of that lattice and its ordinary product image agree in every degree.*

Proof. After the unramified splitting extension, Lemmas 3.1 and 3.2 express every class in the extended graph divisor lattice as a finite sum $D = R_1 + \cdots + R_m$ of rank-one coefficient classes. These are classes in the coefficient extension of the original lattice, not newly descended line bundles. On the elliptic-power source the alternating form of each R_i has rank two, hence $R_i^2 = 0$. Pullback by the isogeny is injective on torsion-free cohomology, so the same identity holds on the quotient. Therefore

$$\frac{D^k}{k!} = \sum_{1 \le i_1 < \cdots < i_k \le m} R_{i_1} \cdots R_{i_k}, \tag{3.10}$$

an O'-linear combination of ordinary products over the splitting ring O'. More precisely,

$$P_{\mathrm{gr}}^k \otimes_O O' = \mathrm{im}\Big(\mathrm{Sym}^k(N_{\mathrm{gr}} \otimes_O O') \longrightarrow H^{2k} \otimes_O O'\Big).$$

Faithful flatness descends the quotient of these finite lattices, and hence descends (3.10) to O, and therefore to $\mathbf{Z}_p$ in the six-axis application. □

The theorem concerns the divisor lattice selected by the marked elliptic-power presentation. If E has no complex multiplication, this is the full Neron–Severi lattice. At a CM fibre extra divisor classes need not satisfy the graph hypotheses. The statement is cohomological: it neither constructs Chow divided powers nor asserts that every integral Hodge class is a divisor product.

For comparison with the classical rational theory, Milne proves that the Lefschetz classes are the rational algebra generated by divisors [25, Theorem 3.2]. Divided powers in Chow and integral Fourier transforms are studied in [26, 3]. The local result above instead keeps the weighted integral graph lattice and the ordinary, rather than divided, product image.

3.2 The six-axis packet

We now apply the graph theorem to the polarization computed in Section 2.

We use the following elementary depth-one lifting observation. Let U be a unimodular O-lattice with symmetric Gram matrix B. Every endomorphism $\bar{T}$ of U/pU that is self-adjoint for $\bar{B}^{-1}$ has a lift $T \in \mathrm{End}_O(U)$ satisfying

$$T^{\mathsf{t}} B^{-1} = B^{-1} T. \tag{3.11}$$

The construction may be performed separately on orthogonal depth summands. Choose any lift T_0. Its adjointness defect has the form

$$T_0^{\mathsf{t}} B^{-1} - B^{-1} T_0 = pD,$$

where D is skew-symmetric with zero diagonal. Choose an integral matrix C with $C^{\mathsf{t}} - C = -D$, for example by assigning its entries strictly above and below the diagonal. Then $T = T_0 + pBC$ reduces to $\bar{T}$, and

$$T^{\mathsf{t}} B^{-1} - B^{-1} T = p(D + C^{\mathsf{t}} - C) = 0.$$

No division by two is used. Applying the construction blockwise preserves the depth decomposition.

Lemma 3.4 (Local chart for the six-axis quotient)**.** *Let $G = 6I_5 - J_5$. At each prime $p \mid 6$, the coefficient lattice has an orthogonal decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where U_0 has rank one and U_1 rank four, both unimodular. The principal quotient is trivial on U_0. On U_1/pU_1, its graph slope has minimal polynomial $t^2 + t + 1$ at $p = 2$; at $p = 3$ it is a scalar, hence a one-point finite-étale block, after choosing a complementary symplectic ruling if the selected half is vertical. In both cases the slope has a depth-preserving integral lift that is self-adjoint for the dual coefficient form, so these data give a marked finite-étale graph presentation.

Proof. Let $e_1, \ldots, e_5$ be the standard basis. Since $\langle e_1, e_1 \rangle = 5$ and $\langle e_1, e_j \rangle = -1$, the vectors

$$v_j = e_j + \frac{1}{5} e_1 \qquad (2 \le j \le 5)$$

span the orthogonal complement of e_1 over $\mathbf{Z}_p$. Its Gram matrix is $\frac{6}{5}(5I_4 - J_4)$, and $\det(5I_4 - J_4) = 5^3$. This gives the asserted unit line and depth-one block at two and three. The unit line contributes nothing to the p-primary discriminant group, so the p-primary isogeny kernel is supported entirely on U_1/pU_1.

At two, Proposition 2.4 applies to every smooth fibre and identifies its geometric kernel with one of the two exotic graphs. Isotropy says that either slope is self-adjoint for the reduced dual form. It is a root in $\mathbf{F}_4$ of $t^2 + t + 1$, so its generated algebra is finite etale. The preceding lifting argument supplies the required integral self-adjoint lift on U_1. At three, the same proposition says that every A_5-stable half is a scalar graph because the coefficient heart is simple with commutant $\mathbf{F}_3$ by Lemma 2.1. More precisely, the four halves form $\mathbf{P}^1(\mathbf{F}_3)$. If the selected member is vertical, choose any other member as the complementary vertical ruling; the selected half then becomes the graph of a scalar (including zero). This change is only in the symplectic ruling of the primary kernel. The chosen symplectic basis of its two-dimensional multiplicity space lifts from $\mathbf{F}_3$ to $\mathbf{Z}_3$: equivalently, reduction $\mathrm{Sp}_2(\mathbf{Z}_3) \to \mathrm{Sp}_2(\mathbf{F}_3)$ is surjective. Thus it lifts to an integral change of ruling preserving the alternating form; the depth-one coefficient block and its scale $P = 3I$ are unchanged. The algebra generated by the resulting scalar is $\mathbf{F}_3$, hence is a one-point finite-étale block, and the same lifting argument applies. Extend each lift by a scalar on U_0. No global choice of either slope is needed. □

Theorem 3.5 (Divided powers for the six-axis family)**.** *For every smooth member X_b of the nonstandard A_5-cubic pencil, the prescribed six-axis graph divisor lattice on J_b has all its integral divided powers in the ordinary divisor-product image. In particular,*

$$\frac{\Theta_b^4}{4!} \in P_{J_b}^4. \tag{3.12}$$

Proof. Only the primes two and three occur in the Smith form $(1, 6, 6, 6, 6)$. Lemma 3.4 verifies the hypotheses of Theorem 3.3 at both primes. At every other prime the isogeny identifies the integral source and target lattices; equivalently the marked presentation has only its unit block $P = I$, so Theorem 3.3 applies trivially (or directly by Lemma 3.2). Thus the image of each desired divided power in the finitely generated quotient $H^{2k}(J_b, \mathbf{Z})/P_{J_b}^k$ vanishes after tensoring with $\mathbf{Z}_p$ for every prime p, and hence vanishes integrally: a finitely generated abelian group splits as a free part and a finite part, and both are detected by these completions. Finally, Lemma 2.3 gives $f_b^* \Theta_b = \lambda_{\mathcal{A}_b}$. The principal polarization is therefore one of the descended graph divisors defining the prescribed lattice, yielding (3.12). Extra Neron–Severi classes at a special fibre can only enlarge the ordinary product image needed for this last assertion. □

Voisin identifies the primitive minimal class as the exact obstruction needed for cubic threefolds.

Corollary 3.6 (Universal CH_0-triviality)**.** *Every smooth member of the nonstandard A_5-cubic pencil is universally CH_0-trivial.*

Proof. By Theorem 3.5, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin's Corollary 4.4 [30] says that, for a smooth complex cubic threefold, this is equivalent to universal CH_0-triviality. □

Two comparisons place the pencil against the parts of the algebraic locus that were known before. Write M_{H_1} for the coarse-moduli image of the pencil, as in Hartlieb's notation. Both comparisons run through one classical criterion.

Lemma 3.7 (Eckardt criterion)**.** *Let $X = \{F = 0\} \subset \mathbf{P}^4$ be a smooth cubic threefold and let $p \in X$. The tangent hyperplane section $T_pX \cap X$ is a cone with vertex p – that is, p is an* Eckardt point *of X – if and only if the Hessian matrix of F at p has rank at most two. Every smooth cubic threefold in either of two classes carries an Eckardt point: those whose defining form is a sum of cubic forms in pairwise disjoint nonempty groups of at most three variables, and the members of the family of [32, Theorem 3.3].*

Proof. Choose coordinates with $p = [1 : 0 : 0 : 0 : 0]$ and write $F = x_0^2L + x_0Q + C$ with L, Q and C in $x_1, \ldots, x_4$. Smoothness at p makes L nonzero, the tangent hyperplane is $\{L = 0\}$, and the section is a cone with vertex p exactly when Q vanishes on $\{L = 0\}$, that is when $Q \in (L)$. The Hessian of F at p is the matrix of the quadric $4x_0L + 2Q$. If $Q = LR$ for a linear form R, that quadric is $L(4x_0 + 2R)$, of rank at most two. Conversely a quadric of rank at most two is a product of two linear forms; the coefficient of x_0^2 vanishes, so at most one factor involves x_0, and if neither did then L would vanish. Comparing the terms linear in x_0 shows that the other factor is proportional to L, so $Q \in (L)$.

Let $F = \sum_j F_j$ be separated, with the j-th form in $n_j \leq 3$ variables and $\sum_j n_j = 5$. Either some $n_j = 2$, or the parts are $3, 1, 1$ or $1, 1, 1, 1, 1$ and at least two of them are singletons. In the first case the binary cubic F_j has a nonzero root, and the corresponding point of $\mathbf{P}^4$, supported on those two variables, lies on X. In the second, let cx^3 and $c'y^3$ be two singleton summands, with c and c' nonzero because X is smooth, and take the point supported on x and y with coordinates satisfying $ca^3 + c'b^3 = 0$. In both cases every diagonal block of the Hessian belonging to another part vanishes there, since a cubic form has vanishing Hessian at the origin, so the rank is at most two.

For the family of [32, Theorem 3.3] work in their coordinates $x_1, \ldots, x_5$ and take p to be the point where $x_1 = 1$ and the other four vanish. It lies on the threefold; every partial derivative of their form vanishes at p except the one in x_3, whose value is their nonzero coefficient t_1, so the tangent hyperplane is $\{x_3 = 0\}$; and the restriction of their form to that hyperplane does not involve x_1, so the section is a cone with vertex p. □

The general member of the pencil carries no such point. That is a computation, recorded in the evidence bundle named in the next proposition: in two independent models of W_5, the sum-zero subspace of the permutation module on the six points of $\mathbf{P}^1(\mathbf{F}_5)$ over $\mathbf{Q}$ and the monomial model induced from a cubic character of a point stabilizer over $\mathbf{Q}(\zeta_3)$, explicit members are smooth with empty Eckardt scheme. The Fermat cubic threefold, which is a member of the pencil and appears explicitly in the monomial model, returns its thirty Eckardt points in both rings as a control.

Proposition 3.8 (Finite Eckardt locus)**.** *Only finitely many points of the coarse-moduli image M_{H_1} of the pencil are represented by smooth cubic threefolds carrying an Eckardt* *point. In particular only finitely many are represented by cubics projectively equivalent to a member of the family of [32, Theorem 3.3].*

Proof. Let B° be the smooth locus of the pencil, as in Theorem 1.5, and F_b the defining form of X_b. Inside $B^\circ \times \mathbf{P}^4$ the locus of pairs (b, p) with $p \in X_b$ and $\operatorname{rank} \operatorname{Hess} F_b(p) \le 2$ is closed, and the projection to B° is proper, so the set of parameters whose member carries such a point is closed. By the computation quoted above it is a proper subset, and B° is a connected curve, so it is finite; M_{H_1} is the image of B°, so finitely many of its points are covered. By Lemma 3.7 those are exactly the points represented by a member carrying an Eckardt point. Every member of the family of [32, Theorem 3.3] carries one, again by Lemma 3.7, and carrying one is invariant under projective equivalence, which gives the second claim. □

Proposition 3.9 (Finite separated-variable intersection)**.** *Only finitely many points of M_{H_1} are represented by smooth cubic threefolds that are projectively equivalent to a cubic whose defining form is a sum of cubic forms in pairwise disjoint nonempty groups of at most three variables. Consequently, all but finitely many points of M_{H_1} are represented by universally CH_0-trivial cubic threefolds by Corollary 3.6, but are not covered by the separated-variable criterion of Colliot-Thélène [11].*

Proof. By Lemma 3.7 a smooth cubic threefold of separated-variable type carries an Eckardt point, and carrying one is invariant under projective equivalence, so Proposition 3.8 leaves only finitely many points of M_{H_1} represented by such cubics. Every remaining point is represented by a universally CH_0-trivial cubic threefold by Corollary 3.6, and by no cubic projectively equivalent to a separated-variable form, so the criterion of [11] does not apply to it. □

The order-three signatures give the same finiteness by a route that uses no computation. Multiplying the variables of a block of size one or two by ζ_3 fixes a separated-variable form, so such a cubic admits an order-three automorphism of signature $(0,0,0,0,1)$ or $(0,0,0,1,1)$; González-Aguilera–Liendo identify these as their families T_3^1 and T_3^2 [13, Theorem 2.5]. Each is the image in moduli of $U^G/N_{\mathrm{GL}_5(\mathbf{C})}(G)$, for U^G the locus of smooth cubics fixed by the relevant group G, under a finite morphism, hence closed [17, Lemmas 2.6–2.7]. The Klein cubic lies in the pencil [17, Lemmas 2.6–2.7 and Proposition 5.7], its automorphism group $\mathrm{PSL}(2, \mathbf{F}_{11})$ has a single conjugacy class of elements of order three, and its signature is $(0,0,1,1,2)$ [13, Proposition 2.6], so it lies in neither family. The pencil is an irreducible curve, so it meets each of the two loci in a finite set. The finiteness of the separated-variable locus therefore does not depend on the registered computation.

All but finitely many moduli points of the pencil escape the separated-variable mechanism of [11] by Proposition 3.9 and the explicit family of [32] by Proposition 3.8, and Proposition 2.5 closes the product route into Voisin's components. None of them asserts that the A_5-curve is disjoint from every known part of Voisin's algebraic-minimal-class locus; the two questions left open there are recorded after Proposition 2.5.

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.6.

4 Hodge atoms and unconditional one stabilization

The proof of Theorem 1.1 uses the ordinary Hodge-atom birational ledger and one new finite invariant on a rank-two atom. The framed formal-monodromy theory is postponed to Section 5; no reconstruction-displacement hypothesis enters this section.

4.1 Ordinary Hodge atoms and conventions

We use the numbered ordinary Hodge-atom theorems of [22] as external inputs. Specifically we use the spectral decomposition of the non-archimedean A-model F-bundle; Definition 5.21 and Proposition 5.22, which associate a well-defined geometric atomic F-bundle equivalence class to a Hodge atom; Proposition 5.23, which makes the underlying Hod-representation an atom invariant; the blowup and projective-bundle chemical formulas used in the proof of Proposition 5.22; Lemma 5.24 for varieties with nef canonical class; and the ordinary non-rationality criterion, Proposition 5.30. The projective-bundle part of that package ultimately rests on the Iritani–Koto decomposition theorem, as explained in [22]. At the time of writing, [22] is an unrefereed preprint; the main theorem inherits that source risk. No enhanced atom, Euler pairing, Serre automorphism, or non-archimedean Riemann–Hilbert assertion from [22] is used here.

We also use Cai's small even quantum connection of a smooth cubic threefold and his curve calculation [7]. The rank-two block data needed below are reproduced explicitly, so the only imported cubic datum is the small even connection matrix itself.

We use one standard rigid-analytic continuation fact. In the form needed below, if Y is a connected normal rigid analytic space and an analytic subset contains a nonempty admissible open, then it is all of Y; this is [9, Lemma 2.1.4]. Smooth rigid spaces are normal.

Lemma 4.1 (The intrinsic Euler endomorphism and the A-model spectral operator)**.** *Let $(\mathcal{H}, \nabla)/B$ be the maximal A-model F-bundle in the conventions displayed in [22]. Let E be the intrinsic Euler field of the associated maximal F-manifold and let*

$$\kappa := \nabla_{u^2\partial_u}|_{u=0}$$

be its residual endomorphism. Then

$$C_E = \kappa, \qquad \kappa = -\mathsf{Eu}\star$$

for the displayed A-model connection $\nabla_{\partial_u} = \partial_u - u^{-2}(\mathsf{Eu}\star) + O(u^{-1})$. Consequently κ and $\mathsf{Eu}\star$ have the same generalized eigenspace decomposition and the same number of distinct eigenvalues; their reduced spectral covers are identified by $\lambda \mapsto -\lambda$. Regularity of the endomorphism and the vanishing locus of its characteristic polynomial discriminant are likewise unchanged.

Proof. By the maximal-F-bundle construction of [22], the intrinsic Euler field is characterized by $C_E = \kappa$. Reading the coefficient of u^{-2} in their displayed A-model connection gives $\kappa = -\mathsf{Eu}\star$. Multiplication of an endomorphism by -1 sends each generalized λ-eigenspace to the generalized $-\lambda$-eigenspace, without changing its dimension or Jordan-block sizes. The remaining claims follow immediately. □

The intrinsic Euler field satisfies $\mathcal{L}_E(\circ) = \circ$ and $[e, E] = e$, and its canonical powers satisfy the Witt identities used below; see [10]. Lemma 4.1 allows us to apply those identities to the intrinsic F-manifold while reading the maximal-eigenvalue locus from the A-model operator $\mathsf{Eu}\star$ used by KKPY. No coordinate formula for the Euler field away from the small point is used.

Lemma 4.2 (The Hodge-fixed base)**.** *Let B_X^{Hod} be the Hodge-fixed locus of the maximal A-model F-bundle of a smooth projective variety X. In the notation of [22, Section 5.2.2], its tangent space is obtained from*

$$H(X)^{\mathsf{Hod}} := \bigoplus_i (H^{i,i}(X) \cap H^{2i}(X, \mathbf{Q}))$$

by extension to the Novikov coefficient field. The restriction of the F-manifold multiplication to TB_X^{Hod} is the corresponding restriction of quantum multiplication. In particular B_X^{Hod} is an F-submanifold and its intrinsic Euler endomorphism is the restriction of κ.

For a smooth cubic threefold, Lefschetz gives

$$H(X)^{\mathsf{Hod}} \otimes \Bbbk = H^{\mathrm{even}}(X) \otimes \Bbbk = \langle 1, P, P^2, P^3 \rangle_{\Bbbk}.$$

Thus B_X^{Hod} is four-dimensional and, at the small hyperplane point, its tangent Euler endomorphism is, up to the sign of Lemma 4.1, Cai's four-dimensional even Euler-multiplication matrix.

Proof. The maximal A-model F-bundle and its multiplication are Hod-equivariant in the construction of [22]. Therefore the fixed tangent subbundle is closed under multiplication, contains the unit, and is preserved by the Euler field; this is the induced F-manifold structure on the smooth fixed locus. For a cubic threefold, $H^{2i}(X, \mathbf{Q}) = \mathbf{Q}P^i$ for $0 \leq i \leq 3$, so the displayed definition of $H(X)^{\mathsf{Hod}}$ is exactly the even cohomology. The final assertion follows from Lemma 4.1. □

4.2 The small-even cubic packet

The following calculation is shared by the atom argument of this section and the framed-monodromy calculation of Section 5. It is entirely a small-connection calculation and is unconditional.

Proposition 4.3 (Small-even cubic block reduction)**.** *Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold, let $P = H|_X$, choose the line class ℓ with $P \cdot \ell = 1$, and put $q = Q^\ell$ and $r = (3q)^{1/2}$. After adjoining r to the Novikov coefficient field, the small even Euler multiplication has characteristic polynomial*

$$T^2(T^2 - 108q),$$

with eigenvalues $6r, -6r, 0, 0$; the zero generalized eigenspace is a single rank-two Jordan block. Moreover a constant coefficient change of basis followed by a unique normalized block-off-diagonal gauge $I + O(z)$, using only integral powers of the original loop coordinate, puts the horizontal system into three blocks

$$\begin{aligned} z^2\partial_z \widetilde{S}_+ &= (6r + O(z^2))\widetilde{S}_+, \\ z^2\partial_z \widetilde{S}_- &= (-6r + O(z^2))\widetilde{S}_-, \\ z^2\partial_z \begin{pmatrix} \widetilde{S}_3 \\ \widetilde{S}_4 \end{pmatrix} &= \left[J_0 + zD_0 + z^2E_0 + O(z^3)\right] \begin{pmatrix} \widetilde{S}_3 \\ \widetilde{S}_4 \end{pmatrix}, \end{aligned} \tag{4.1}$$

where

$$J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad D_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \qquad E_0 = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}. \tag{4.2}$$

Proof. Cai's small even horizontal system, in the ordered classical basis $1, P, P^2, P^3$, is

$$\begin{aligned} z^2\partial_z S &= (K_X + zG_X)S, \\ K_X &= 2\begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \\ G_X &= \frac{1}{2}\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}. \end{aligned} \tag{4.3}$$

This is the displayed system in [7, Section 3, p. 4].[1] The constant change of basis

$$C = \begin{pmatrix} 6r^3 & -6r^3 & 0 & -7r^2 \\ 7r^2 & 7r^2 & -2r^2 & 0 \\ 3r & -3r & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \qquad \det C = -486r^5, \qquad S = C\widehat{S}, \tag{4.4}$$

puts the constant term into

$$C^{-1}K_X C = \begin{pmatrix} 6r & 0 & 0 & 0 \\ 0 & -6r & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{4.5}$$

In particular its characteristic and minimal polynomials are both $T^2(T^2 - 108q)$.

Partition the coordinates as $1 \mid 1 \mid 2$. There is a unique formal gauge

$$A(z) = I + \sum_{n\geq 1} A_n z^n, \qquad A_n \in \mathrm{Mat}_4(\mathbf{C}[r, r^{-1}]), \tag{4.6}$$

whose positive coefficients are block-off-diagonal and for which $\widehat{S} = A(z)\widetilde{S}$ makes every coefficient block diagonal. At order n the required coefficient is the unique solution of a Sylvester equation between blocks whose scalar spectra are distinct; the nilpotent part of the zero block is only a nilpotent perturbation of the invertible Sylvester operator. Thus the gauge and its inverse use integral powers of z. Direct multiplication gives exactly (4.1) and (4.2). This agrees with Cai's reduction [7, Section 3, pp. 4–5]. □

4.3 The cubic zero packet is an atom

We first remove a point that cannot safely be delegated to a worked example: the double zero eigenvalue at the small hyperplane point must remain a repeated eigenvalue on the Hodge base, so that this point lies in the maximal-eigenvalue locus used to define atoms.

4.3.1 A discriminant lemma for a regular four-dimensional F-manifold

Let $(M, \circ, e, E)$ be an F-manifold over a characteristic-zero field. Put

$$U = C_E : T_M \longrightarrow T_M, \qquad X_s = E^{\circ s}, \quad X_0 = e.$$

[1] In arXiv:2608.01577v1, one isolated prose line states the two nonzero Euler eigenvalues with a factor two smaller. We follow Cai's displayed matrix, its transformed block, and the subsequent scalar equations, which are mutually consistent and give the values $\pm 6r$ used here.

At a point where U is regular, $X_0, \dots, X_{n-1}$ form a local frame, where $n = \dim M$. The canonical frame satisfies

$$[X_i, X_j] = (j - i)X_{i+j-1}. \tag{4.7}$$

These are the standard Witt identities for a regular F-manifold; see [10]. We need only the case $n = 4$.

Write

$$P(T) = \det(TI - U) = T^4 + \lambda_3 T^3 + \lambda_2 T^2 + \lambda_1 T + \lambda_0 \tag{4.8}$$

and let $\Delta = \operatorname{disc}(P)$.

Lemma 4.4 (Discriminant differential equation)**.** *Assume U is regular at $p \in M$. On the germ of M at p,*

$$X_0(\Delta) = 0, \tag{4.9}$$

$$X_1(\Delta) = 12\Delta, \tag{4.10}$$

$$X_2(\Delta) = -6\lambda_3\Delta, \tag{4.11}$$

$$X_3(\Delta) = (6\lambda_3^2 - 10\lambda_2)\Delta. \tag{4.12}$$

Consequently there is a regular one-form ω on the germ such that

$$d\Delta = \Delta\,\omega. \tag{4.13}$$

If $\Delta(p) = 0$, then Δ vanishes identically on the germ at p.

Proof. Cayley–Hamilton, applied to the unit, is the vector-field relation

$$X_4 + \lambda_3 X_3 + \lambda_2 X_2 + \lambda_1 X_1 + \lambda_0 X_0 = 0. \tag{4.14}$$

Taking Lie brackets of (4.14) with X_s, using (4.7), reducing every X_m with $m \geq 4$ by (4.14) and its products by E, and comparing coefficients in the frame $X_0, \dots, X_3$ gives the following coefficient identities:

$$X_0(\lambda_k) = -(k+1)\lambda_{k+1}, \tag{4.15}$$

$$X_1(\lambda_k) = (4 - k)\lambda_k, \tag{4.16}$$

$$X_2(\lambda_k) = -\lambda_3\lambda_k + (5 - k)\lambda_{k-1}, \tag{4.17}$$

$$X_3(\lambda_k) = (\lambda_3^2 - 2\lambda_2)\lambda_k + (6 - k)\lambda_{k-2} - \lambda_3\lambda_{k-1}, \tag{4.18}$$

for $0 \leq k \leq 3$, with the conventions $\lambda_4 = 1$ and $\lambda_j = 0$ for $j < 0$. These are the $n = 4$ cases of the coefficient identities in [10, (19),(20),(24),(25)]; the calculation above shows that only the Witt identities and Cayley–Hamilton enter.

For completeness, we now compute the action on the discriminant without any appeal to a splitting theorem on M. Work universally in the polynomial ring in formal roots $\mu_1, \dots, \mu_4$, so that

$$P(T) = \prod_{i=1}^{4}(T - \mu_i).$$

The derivations of the coefficient ring specified by (4.15)–(4.18) are the restrictions of

$$D_s = \sum_{i=1}^{4} \mu_i^s \frac{\partial}{\partial \mu_i}, \qquad 0 \leq s \leq 3.$$

This is checked directly on the elementary symmetric functions, and is exactly equivalent to (4.15)–(4.18). Since

$$\Delta = \prod_{i<j} (\mu_i - \mu_j)^2,$$

we obtain

$$\frac{D_s \Delta}{\Delta} = 2 \sum_{i<j} \frac{\mu_i^s - \mu_j^s}{\mu_i - \mu_j}. \tag{4.19}$$

The right side is symmetric, hence is a polynomial in the λ_k. For $s = 0, 1, 2, 3$ it equals respectively

$$0, \qquad 12, \qquad -6\lambda_3, \qquad 6\lambda_3^2 - 10\lambda_2.$$

Indeed, the $s = 2$ quotient is $\mu_i + \mu_j$, and the $s = 3$ quotient is $\mu_i^2 + \mu_i\mu_j + \mu_j^2$; summing over pairs and using $\sum_i \mu_i = -\lambda_3$ and $\sum_{i<j} \mu_i\mu_j = \lambda_2$ gives the displayed expressions. This proves (4.9)–(4.12). Since $X_0, \ldots, X_3$ form a frame, these equations are equivalent to (4.13) for a regular one-form ω.

It remains to prove the vanishing statement in the non-archimedean analytic germ. Let $\mathfrak{m}_p$ be the maximal ideal of the local ring. Passing to the completed local ring at the rigid point p, choose regular parameters $y_1, \ldots, y_4$; smoothness gives a noncanonical isomorphism

$$\widehat{\mathcal{O}}_{M,p} \simeq \kappa(p)[[y_1, \ldots, y_4]],$$

where $\kappa(p)$ is the residue field of p. (In the cubic application below, $p = b$ is $\Bbbk$-rational.) Suppose $\Delta \neq 0$ there. Since $\Delta(p) = 0$, let Δ_m be its lowest nonzero homogeneous part, of degree $m \geq 1$. In $d\Delta = \Delta\omega$, the left side has lowest possible degree $m - 1$, whereas the right side has degree at least m. Hence $d\Delta_m = 0$. Characteristic zero implies that all partial derivatives of Δ_m vanish, so Euler's identity gives $m\Delta_m = \sum y_i \partial_{y_i} \Delta_m = 0$, a contradiction. Thus $\Delta = 0$ in the completion. Since the local ring is Noetherian, Krull's intersection theorem makes the map to its completion injective, so $\Delta = 0$ as an analytic germ. □

4.3.2 Application to a cubic threefold

Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold and use the notation of Proposition 4.3. That proposition gives the small even characteristic polynomial

$$T^2(T^2 - 108q) \tag{4.20}$$

and the eigenvalues

$$6r, \quad -6r, \quad 0, \quad 0, \tag{4.21}$$

with a single rank-two Jordan block at zero. By Lemma 4.1, the intrinsic F-manifold Euler endomorphism is this operator up to an overall sign. In either sign the minimal polynomial equals the characteristic polynomial, so the Euler endomorphism is regular at the small hyperplane point b, and the quartic discriminant has the same zero locus.

By Lemma 4.2, the Hodge-fixed base B_X^{Hod} is a four-dimensional F-submanifold and its tangent Euler endomorphism is exactly the four-dimensional operator to which Lemma 4.4 applies (up to the harmless overall sign of Lemma 4.1). Since the quartic (4.20) has a repeated root, its discriminant vanishes at b. Hence it vanishes on an admissible neighborhood of b.

KKPY prove that B_X^{Hod} is connected and smooth. It is therefore connected and normal. The zero locus of the discriminant is an analytic subset containing a nonempty admissible open, so [9, Lemma 2.1.4] gives

$$\Delta \equiv 0 \qquad \text{on } B_X^{\mathsf{Hod}}. \tag{4.22}$$

Thus the even Euler endomorphism has at most three distinct eigenvalues at every point, while (4.21) shows that it has exactly three at b.

We must compare this count with the spectrum on all of $H^\bullet(X)$, since KKPY define the maximal-eigenvalue locus using the full F-bundle.

Lemma 4.5 (Spectrum transfer)**.** *At every even point of the Hodge base, the set of eigenvalues of Euler multiplication on $H^\bullet(X)$ is the same as the set of eigenvalues on $H^{\mathrm{even}}(X)$.*

Proof. The even quantum cohomology $A = H^{\mathrm{even}}(X)$ is a finite-dimensional commutative algebra over the coefficient field, and $H^\bullet(X)$ is a unital A-module because the bulk point is even. Let $A = \bigoplus_i A_i$ be the generalized-eigenvalue decomposition for multiplication by the Euler element, with corresponding orthogonal idempotents ε_i. Then

$$H^\bullet(X) = \bigoplus_i \varepsilon_i H^\bullet(X).$$

Each summand is nonzero, since $\varepsilon_i = \varepsilon_i \cdot 1 \in \varepsilon_i H^\bullet(X)$. On the local algebra A_i, the element $E - \lambda_i$ is nilpotent, so its action on every finite A_i-module is nilpotent. Hence the only eigenvalue on $\varepsilon_i H^\bullet(X)$ is λ_i. Every even eigenvalue therefore occurs on the full module, and no additional eigenvalue occurs there. □

It follows from (4.22) and Lemma 4.5 that the number of eigenvalues on the full cohomology is everywhere at most three and is exactly three at b. Therefore b lies in the open locus U_X of KKPY where that number is globally maximal.

The generalized eigenbundle ranks are locally constant on the unramified reduced spectral cover and hence constant on each connected component. At b the two nonzero branches in (4.21) have rank one. The zero branch has even rank two. Its odd rank is ten: indeed $b_3(X) = 10$, since

$$c(TX) = \frac{(1+P)^5}{1+3P} = 1 + 2P + 4P^2 - 2P^3,$$

so $\chi_{\mathrm{top}}(X) = \int_X c_3(TX) = -6$, while Lefschetz and Poincaré duality give $b_0 = b_2 = b_4 = b_6 = 1$ and $b_1 = b_5 = 0$. Thus $4 - b_3 = -6$. Moreover Euler multiplication annihilates $H^3(X)$ at b. In Novikov degree $d \geq 0$, multiplication by $c_1(X)$ sends a class of degree 3 to cohomological degree

$$3 + 2 - 2c_1(X) \cdot d = 5 - 4d.$$

For $d = 0$ this is $H^5(X) = 0$, for $d = 1$ it is $H^1(X) = 0$, and for $d \geq 2$ it is negative. Hence the entire $H^3(X)$ lies in the zero packet.

We have therefore proved the following without using KKPY's worked cubic example.

Proposition 4.6 (The cubic zero atom)**.** *For every smooth cubic threefold X, the small hyperplane point $b \in B_X^{\mathsf{Hod}}$ lies in the maximal-eigenvalue locus U_X. Over b the unramified reduced spectral cover has two rank-one branches corresponding to the two nonzero eigenvalues and one branch whose generalized eigenbundle has parity ranks*

$$(\mathrm{rk}_{\bar{0}}, \mathrm{rk}_{\bar{1}}) = (2, 10). \tag{4.23}$$

The rank-twelve branch is a connected component of the cover and hence defines a Hodge atom, denoted α_X.

Proof. Only the last assertion remains. Over U_X the reduced spectral cover is finite etale. On each connected component of that cover, the corresponding generalized-eigenbundle is locally free of constant rank. The fibre over b consists of three spectral points: the point belonging to the zero packet has generalized-eigenbundle rank 12, while the two points belonging to the nonzero eigenvalues have rank 1. Hence neither rank-one point can lie on the connected component through the rank-twelve point. The fibre of that component over b therefore consists only of the zero spectral point; in particular the component is the atomic component determined by the zero packet (and has degree one over its image in a neighborhood of b). Its parity ranks are (4.23). $\square$

By KKPY's invariance of the Hod-representation of an atom, the parity ranks (4.23) are invariants of the global Hodge atom α_X.

4.4 A rank-two atomic residue discriminant

We now construct the invariant that distinguishes α_X from every low-dimensional atom that could have the same parity ranks.

4.4.1 Pairing and spectral factors

Let $(\mathcal{H}, \nabla)/B$ be the non-archimedean A-model F-bundle of a smooth projective variety, restricted to the purely even Hodge-fixed base. In the non-archimedean construction of [22, Section 3.5.2], the inverse Gram matrix of the Poincaré form is a constant analytic section PD^{-1} and is used to define quantum multiplication. Thus the Poincaré form itself is available on this analytic F-bundle, constant in the standard cohomology frame. We prove its horizontality below directly and coefficientwise, rather than importing the later convergent complex-analytic discussion of the pairing. Define the u-sesquilinear pairing

$$\psi_u(s, t) = (s(u), t(-u))_{\mathrm{Poinc}}.$$

Lemma 4.7 (Algebraic horizontality)**.** *The pairing ψ is horizontal for the non-archimedean quantum connection. No convergence assumption is required.*

Proof. Quantum multiplication is Frobenius:

$$(a \star b, c) = (a, b \star c).$$

Thus every quantum multiplication operator, in particular Euler multiplication and multiplication by an even tangent vector, is self-adjoint for the Poincaré pairing. The grading operator

$$\mathsf{Gr}|_{H^a} = \frac{a - \dim X}{2}\,\mathrm{id}$$

is anti-self-adjoint because Poincaré duality pairs H^a with $H^{2\dim X - a}$. Substituting these two identities into the displayed quantum-connection formulas proves horizontality coefficientwise in the Novikov and bulk variables. The proof is algebraic and therefore takes place directly over the non-archimedean coefficient ring. $\square$

The even part is a sub-F-bundle: the Hodge-fixed base is purely even and quantum multiplication by an even class, the Euler operator, and the grading operator all preserve cohomological parity.

We also need that the pairing restricts nondegenerately to a spectral factor, without assuming an isometric block-splitting gauge.

Lemma 4.8 (Orthogonality of separated spectral factors)**.** *Suppose locally the F-bundle is split into factors whose leading Euler eigenvalues λ_i are pairwise distinct. Then the horizontal Poincaré pairing is block diagonal for that splitting; in particular its restriction to each factor, and to the even part of each factor, is nondegenerate.*

Proof. Choose any regular block-diagonalizing frame for the connection and write the pairing matrix as $P(u) = \sum_{m\ge 0} P_m u^m$. On the (i,j) off-diagonal block, the coefficient of u^{-1} in horizontality is

$$U_i^T (P_0)_{ij} - (P_0)_{ij} U_j = 0.$$

The Sylvester operator

$$X \longmapsto U_i^T X - X U_j$$

is invertible because $\lambda_i - \lambda_j$ is a unit and the remaining parts of U_i, U_j are nilpotent. Hence $(P_0)_{ij} = 0$. Inductively, the coefficient of each successive power of u has the same invertible Sylvester operator on $(P_m)_{ij}$ and a right-hand side involving only earlier off-diagonal coefficients, already zero. Thus $P_{ij}(u) = 0$ identically. Since the full pairing is nondegenerate, every diagonal restriction is nondegenerate. The Poincaré form pairs only equal parity, so the even restriction is also nondegenerate. □

4.4.2 The elementary modification

Let $\mathcal{U}$ be a connected component of the unramified spectral cover representing a geometric Hodge atom, and suppose the even part E of its geometric atomic F-bundle has rank two. Let U be its leading Euler operator. On the spectral component U has a single eigenvalue λ; put

$$N = \mathsf{U} - \lambda I, \qquad \lambda = \frac{1}{2}\operatorname{tr}(\mathsf{U}). \tag{4.24}$$

Then $N^2 = 0$. Assume N is generically nonzero on $\mathcal{U}$.

Center the scalar irregular part algebraically. If the original matrices are A^{old} and B_δ^{old}, replace them by

$$A = A^{\mathrm{old}} - u^{-1}\lambda I, \qquad B_\delta = B_\delta^{\mathrm{old}} + u^{-1}(\delta\lambda) I.$$

This is the connection obtained formally by multiplying horizontal sections by $\exp(\lambda/u)$, but no exponential is adjoined and no convergence is being asserted. The added scalar one-form is exact, so flatness is unchanged. In a local frame horizontal sections for the centered connection satisfy

$$u\partial_u y = A(u)y, \qquad \delta y = B_\delta(u)y, \tag{4.25}$$

with

$$A(u) = u^{-1}N + A_0 + uA_1 + u^2 A_2 + \cdots, \tag{4.26}$$

$$B_\delta(u) = u^{-1}C_\delta + C_{\delta,0} + uC_{\delta,1} + \cdots. \tag{4.27}$$

The centering does not alter the pairing identity: the scalar terms in the u and $-u$ factors have opposite signs and cancel. Flatness is

$$\delta A - u\partial_u B_\delta + [A, B_\delta] = 0. \tag{4.28}$$

On the open locus $\mathcal{U}^\times = \{N \ne 0\}$,

$$L := \operatorname{im} N = \ker N$$

is a line.

Lemma 4.9 (The regular coefficient preserves the nilpotent line)**.** *On $\mathcal{U}^\times$,*

$$A_0 L \subset L. \tag{4.29}$$

Proof. By Lemmas 4.7 and 4.8, the rank-two even factor has a horizontal nondegenerate pairing

$$P(u) = P_0 + uP_1 + u^2 P_2 + \cdots ,$$

with P_0 symmetric. Horizontality in the u-direction reads

$$u\partial_u P(u) + A(u)^T P(u) + P(u)A(-u) = 0. \tag{4.30}$$

The u^{-1} coefficient gives

$$N^T P_0 = P_0 N. \tag{4.31}$$

Hence N is self-adjoint for P_0. Since $N^2 = 0$,

$$(Nx, Ny) = (x, N^2 y) = 0,$$

so $L = \operatorname{im} N$ is isotropic. In a nondegenerate two-dimensional symmetric space, $L^\perp = L$.

The constant coefficient of (4.30) is

$$A_0^T P_0 + P_0 A_0 + N^T P_1 - P_1 N = 0. \tag{4.32}$$

For $0 \neq x \in L = \ker N$, sandwiching (4.32) between x on both sides kills the two terms containing N and gives $2(A_0 x, x) = 0$. Thus $A_0 x \in x^\perp = L^\perp = L$, proving (4.29). □

Define the elementary modification

$$E^\sharp = \{s \in E : s \bmod u \in L\}. \tag{4.33}$$

This is intrinsic. Choose locally a frame e_1, e_2 with

$$L = \langle e_1 \rangle, \qquad N e_1 = 0, \qquad N e_2 = \nu e_1, \qquad \nu \in \mathcal{O}^\times. \tag{4.34}$$

Then $E^\sharp$ has frame e_1, ue_2, corresponding to $S = \operatorname{diag}(1, u)$. The transformed u-matrix is

$$A^\sharp = S^{-1} A S - S^{-1} u \partial_u S. \tag{4.35}$$

Because

$$S^{-1} E_{12} S = u E_{12}, \qquad S^{-1} E_{21} S = u^{-1} E_{21},$$

the term $u^{-1} N = \nu u^{-1} E_{12}$ becomes the regular term νE_{12}. The only possible pole coming from A_0 is its E_{21} coefficient, and Lemma 4.9 says that coefficient vanishes. Every $u^m A_m$, $m \geq 1$, remains regular after losing at most one power of u. Hence

$$A^\sharp(u) = R + uR_1 + u^2 R_2 + \cdots . \tag{4.36}$$

The conjugacy class of the residue R is intrinsic to the modified lattice.

We use only its discriminant

$$\delta^\sharp := (\operatorname{tr} R)^2 - 4 \det R. \tag{4.37}$$

This is invariant under conjugacy and also under adding a scalar multiple of the identity to R. The latter observation makes the invariant insensitive to harmless scalar residue normalizations.

4.4.3 Flatness freezes the residue discriminant

Proposition 4.10 (Rank-two rigidity)**.** *On $\mathcal{U}^\times$, the modified connection is regular in every base direction and the residue satisfies*

$$\delta R = [G_\delta, R] \tag{4.38}$$

for a regular matrix G_δ. Consequently

$$\delta(\delta^\sharp) = 0 \tag{4.39}$$

for every base vector field δ.

Proof. The u^{-2} coefficient of (4.28) is

$$[N, C_\delta] = 0. \tag{4.40}$$

The u^{-1} coefficient is

$$\delta N + C_\delta + [N, C_{\delta,0}] + [A_0, C_\delta] = 0. \tag{4.41}$$

Since $\operatorname{tr} N = 0$, taking traces gives $\operatorname{tr} C_\delta = 0$. On $\mathcal{U}^\times$ the commutant of a nonzero rank-one nilpotent 2×2 matrix is $\mathcal{O}I \oplus \mathcal{O}N$. Equations (4.40) and $\operatorname{tr} C_\delta = 0$ therefore give

$$C_\delta = q_\delta N. \tag{4.42}$$

Transform the base equation to the modified lattice. In the moving adapted frame (4.34), the frame-connection contribution is included in $C_{\delta,0}$; the matrix $S = \operatorname{diag}(1, u)$ itself is base-independent. The leading term $u^{-1} q_\delta N$ becomes regular. Thus the only possible pole has the form

$$B_\delta^\sharp = u^{-1} K_\delta + G_\delta + O(u), \qquad K_\delta = k_\delta E_{21}. \tag{4.43}$$

Flatness in the modified lattice gives at order u^{-1}

$$K_\delta + [R, K_\delta] = 0. \tag{4.44}$$

The $(1,2)$ entry of R is exactly the unit ν coming from N, so write

$$R = \begin{pmatrix} a & \nu \\ c & d \end{pmatrix}.$$

Then

$$[R, k_\delta E_{21}] = k_\delta \begin{pmatrix} \nu & 0 \\ d - a & -\nu \end{pmatrix}.$$

The two diagonal entries of (4.44) are therefore νk_δ and $-\nu k_\delta$. Since ν is a unit, $K_\delta = 0$. Hence the base connection is regular after the modification. The constant coefficient of flatness is now

$$\delta R + [R, G_\delta] = 0,$$

which is (4.38). Trace and determinant are invariant under infinitesimal conjugation, so (4.39) follows. In particular $\delta^\sharp$ is locally constant on $\mathcal{U}^\times$: on a smooth rigid chart with regular parameters $t_1, \dots, t_m$, apply (4.39) to the coordinate derivations. In the completed local ring every positive-degree coefficient of the resulting power series is then zero, so the function is constant on each connected chart. □

Proposition 4.11 (Exponent interpretation of the residue discriminant)**.** *Fix a point of $\mathcal{U}^\times$. Let K be its coefficient field and work in an algebraic closure $\overline{K}$ over which the residue R splits. Let ρ_1, ρ_2 be the eigenvalues of the residue of the canonical modified lattice $E^\sharp$, counted with algebraic multiplicity. Then:*

(i) the classes $[\rho_1], [\rho_2] \in \overline{K}/\mathbf{Z}$ are the formal exponent classes of the original centered *rank-two differential module; and*

(ii)

$$\delta^\sharp = (\rho_1 - \rho_2)^2. \tag{4.45}$$

Thus $\delta^\sharp$ records the squared separation of the residue representatives selected by the canonical elementary modification. It is invariant under permuting the two representatives and under a common scalar shift. It is not, in general, determined by the two exponent classes modulo $\mathbf{Z}$ *alone.*

Proof. The elementary modification is intrinsic, and in an adapted frame it is implemented by $S = \operatorname{diag}(1, u)$ (or by $\operatorname{diag}(u, 1)$ after reversing the orientation of the nilpotent Jordan block). In either form $S \in \mathrm{GL}_2(\overline{K}((u)))$ and uses only integral powers of the original loop coordinate. Hence it is a meromorphic gauge isomorphism of differential modules; because this particular gauge multiplies the adapted basis vectors only by integral powers of u, it changes the corresponding exponent representatives only by integers and therefore preserves their classes in $\overline{K}/\mathbf{Z}$.

By Proposition 4.10, the modified connection is regular singular and has the form

$$u\partial_u y^\sharp = (R + O(u))y^\sharp.$$

For a regular-singular formal differential module, the formal exponent classes are the residue eigenvalues modulo $\mathbf{Z}$, including in the resonant or nonsemisimple case; see, for example, the regular-singular part of the formal classification in [27, Chapter 3]. Therefore $[\rho_1], [\rho_2]$ are the formal exponent classes of the original centered block.

Finally,

$$(\operatorname{tr} R)^2 - 4 \det R = (\rho_1 + \rho_2)^2 - 4\rho_1\rho_2 = (\rho_1 - \rho_2)^2,$$

which proves (4.45). $\square$

4.4.4 Crossing the locus where N vanishes

The preceding calculation was made only where $N \neq 0$. That is enough. We first record the gluing point implicit in the spectral decomposition.

Lemma 4.12 (Gluing of a separated spectral factor)**.** *Over the unramified locus U_X, the local F-bundle factors belonging to a fixed connected component $\mathcal{U}$ of the reduced spectral cover glue, up to regular block-diagonal change of frame. In particular any regular-gauge-invariant meromorphic expression formed from the factor is a well-defined meromorphic function on $\mathcal{U}$.*

Proof. At $u = 0$, the proof of [22, Proposition 5.23] constructs the generalized-eigenspace subbundle over U_X associated with each component of the reduced spectral cover, and the spectral decomposition theorem of [22] supplies the corresponding local analytic F-bundle factors. Thus the factor itself is part of the ordinary atom package. We record only the uniqueness needed to glue the higher u-jet expressions used below.

Fix the splitting at $u = 0$. Among regular gauges $g = I + O(u)$ carrying the connection to block-diagonal form, impose the normalization that every positive u-coefficient of g is block-off-diagonal. At each order the unknown off-diagonal coefficient is determined by a Sylvester operator of the form

$$X \longmapsto U_i X - X U_j.$$

It is invertible because the scalar parts of U_i and U_j are distinct on U_X; the remaining nilpotent terms give only a nilpotent perturbation of multiplication by their unit difference. Hence a normalized splitting, if chosen, is unique. The analytic splittings supplied by KKPY therefore agree on overlaps after a regular block-diagonal change of frame. Consequently regular-gauge-invariant meromorphic expressions formed from one spectral factor glue along its connected component $\mathcal{U}$. □

Lemma 4.13 (Meromorphic continuation on a spectral component)**.** *Assume N is generically nonzero on the connected spectral component $\mathcal{U}$. Then $\delta^\sharp$ extends as a meromorphic function on $\mathcal{U}$ and is constant there. In particular it is independent of the chosen base point representing the same local atom.*

Proof. The reduced spectral cover is unramified over the smooth Hodge base on U_X. Hence $\mathcal{U}$ is smooth and therefore normal. A connected normal rigid space is irreducible by the rigid-analytic irreducibility theory of [9].

By Lemma 4.12, the rank-two factor is defined along $\mathcal{U}$ up to regular gauge. Over its meromorphic function field, the nonzero nilpotent matrix N has rank one and its image line is generated by any nonzero column. Choose a meromorphic adapted frame. In that frame the coefficients of the connection and therefore the residue R of the modified lattice are meromorphic. Since (4.37) is gauge invariant, $\delta^\sharp$ is a meromorphic function on $\mathcal{U}$.

Choose a nonempty admissible open in $\mathcal{U}^\times$. By Proposition 4.10, $\delta^\sharp$ is constant on a smaller connected open there; call the value c. The difference $\delta^\sharp - c$ is an element of the meromorphic function field of the integral rigid space $\mathcal{U}$. To spell out the identity principle, choose a connected normal affinoid subdomain meeting that open and write the difference there as a/b with a, b analytic and b nonzero. On a smaller nonempty admissible open where b is invertible the quotient vanishes, so a vanishes there. Conrad's identity principle for connected normal rigid spaces gives $a = 0$ on the affinoid. Since a meromorphic function on the irreducible space $\mathcal{U}$ is determined by its value on any nonempty admissible open, $\delta^\sharp - c = 0$ in its meromorphic function field. In particular the same value is obtained on every component of $\mathcal{U}^\times$, even if those components are separated by points at which $N = 0$. □

4.4.5 Descent to ordinary Hodge atoms

Proposition 4.14 (Atomic residue discriminant)**.** *Let α be an ordinary Hodge atom whose associated geometric atomic F-bundle has even rank two and whose centered leading operator is generically nonzero nilpotent. Then the number $\delta^\sharp(\alpha)$ defined by* (4.37) *is a well-defined invariant of the ordinary Hodge atom.*

Proof. First consider an isomorphism of geometric atomic F-bundles. It preserves cohomological parity, hence preserves the even rank. Its value at $u = 0$ conjugates the centered leading operator N to N'; consequently generic nonvanishing of N is preserved, and the isomorphism carries $L = \operatorname{im} N$ to $L' = \operatorname{im} N'$. Hence it carries the intrinsic elementary modification $E^\sharp$ to $(E')^\sharp$. The residues are conjugate after applying the same trace-centering on both sides. Any residual scalar shift of R is in any event killed by (4.37); therefore both the domain condition and $\delta^\sharp$ are preserved.

Second, changing the base point within one connected component of the unramified spectral cover does not change the even rank or the property that N is generically nonzero on that component, and it preserves $\delta^\sharp$ by Lemma 4.13. Thus the domain of definition and the value of $\delta^\sharp$ are invariant under the two equivalences used by KKPY for geometric atomic F-bundles.

There are two further quotients in the passage to global Hodge atoms. First, KKPY quotient components attached to a fixed variety by $\mathrm{Aut}(X)$. An automorphism pulls back the A-model F-bundle and carries the relevant spectral component to the corresponding one, so the isomorphism case above shows that both the domain condition and $\delta^\sharp$ are unchanged. Second, HAtoms is further quotiented by the elementary relations coming from disjoint unions, blowups with smooth centers, and projective bundles. Proposition 5.22 of [22] proves exactly that the map from these local atom classes to geometric atomic F-bundle equivalence classes factors through those elementary relations: the two sides of each relation have the same image in the target set of geometric atomic F-bundle classes. Since the rank-two/generic-nilpotence domain condition and $\delta^\sharp$ have already been shown to be well defined on that target set, composing with the map of Proposition 5.22 gives a well-defined invariant on precisely the ordinary Hodge-atom classes satisfying the stated domain condition. ☐

Remark 4.15. The Poincaré pairing is used only as an auxiliary structure to prove the intrinsic property (4.29) of the unenhanced atomic connection. No pairing is added to the definition of the atom, and morphisms of atoms are not required to preserve an additional enhancement.

4.5 The cubic residue and low-dimensional exclusion

4.5.1 The residue discriminant of the cubic atom

For the specialization of the abstract rank-two construction above to Proposition 4.3, we identify its formal loop coordinate u with the original loop coordinate z used in the cubic small-even system. By Proposition 4.3, the rank-two zero block is

$$z^2\partial_z Y = (J_0 + zD_0 + z^2E_0 + O(z^3))Y$$

with the matrices in (4.2). Dividing by z, the centered leading term is $N = J_0 = 2E_{12}$ and $L = \langle e_1\rangle$. The canonical elementary modification uses $S = \mathrm{diag}(1, z)$, so Formula (4.35) gives the residue

$$R_X = 2E_{12} + D_0 - \frac{8}{81}E_{21} - \mathrm{diag}(0,1) = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}. \tag{4.46}$$

Hence

$$\mathrm{tr}\, R_X = -1, \qquad \det R_X = \frac{5}{36},$$

and its residue characteristic polynomial is

$$T^2 + T + \frac{5}{36} = \left(T + \frac{1}{6}\right)\left(T + \frac{5}{6}\right).$$

Thus the canonical modified-residue representatives are $\rho_1 = -1/6$ and $\rho_2 = -5/6$. Proposition 4.11 therefore gives

$$\boxed{\delta^\sharp(\alpha_X) = \left(-\frac{1}{6} + \frac{5}{6}\right)^2 = \frac{4}{9}.} \tag{4.47}$$

The occurrence of $(E_0)_{21} = -8/81$ in R_X is important: the modification retains precisely the next coefficient that controls this formal exponent separation.

4.5.2 No curve represents the cubic atom

Suppose α_X is represented by a smooth projective curve C. If $C = \mathbf{P}^1$, KKPY's projective-bundle formula gives

$$\mathsf{CF}(\mathbf{P}^1) = 2\mathsf{CF}(\mathrm{pt}),$$

so only point atoms occur; their rank is one, whereas α_X has rank 12.

Assume $g(C) \geq 1$. Then K_C is nef, so KKPY's nef-canonical lemma says that C has a single Hodge atom, carrying all of $H^\bullet(C)$. Its parity ranks are $(2, 2g)$. By the Hodge-representation invariance of atoms and (4.23), an identification with α_X would force

$$g = 5. \tag{4.48}$$

We now exclude that possibility by the residue discriminant.

Let $p \in H^2(C)$ be normalized by $\int_C p = 1$ and put $a = 2 - 2g$. For $g \geq 1$ there are no nonconstant genus-zero maps $\mathbf{P}^1 \to C$, so the big genus-zero quantum product is the classical cup product. On the even basis $1, p$, after removing the scalar H^0-direction, Cai's connection is

$$u\partial_u Y = \left[u^{-1} a E_{21} + \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix}\right] Y; \tag{4.49}$$

see [7, §4]. For $g > 1$, $a \neq 0$ and $N = aE_{21}$, so $L = \langle e_2 \rangle$. The elementary modification now uses $S = \mathrm{diag}(u, 1)$ and gives

$$R_C = aE_{21} + \mathrm{diag}(1/2, -1/2) - \mathrm{diag}(1, 0) = aE_{21} - \frac{1}{2}I. \tag{4.50}$$

The modified residue has the repeated eigenvalue $-1/2$, so its two canonical residue representatives coincide. By Proposition 4.11,

$$\boxed{\delta^\sharp(C) = \left(-\frac{1}{2} + \frac{1}{2}\right)^2 = 0.} \tag{4.51}$$

Equivalently, its modified residue characteristic polynomial is $(T + 1/2)^2$.

For the only possible competitor $g = 5$, equations (4.47) and (4.51) contradict the well-definedness of $\delta^\sharp$ on ordinary Hodge atoms. We have proved:

Proposition 4.16. *The cubic atom α_X is not represented by any smooth projective variety of dimension at most one.*

4.5.3 No surface represents the cubic atom

Proposition 4.17. *The atom α_X is not represented by a smooth projective surface. Together with Proposition 4.16 and additivity for disjoint unions, this excludes every smooth projective variety of dimension at most two. Consequently*

$$\alpha_X \notin \mathsf{HAtoms}_{\dim \leq 2}. \tag{4.52}$$

Proof. Suppose α_X occurs in the atomic composition of a smooth projective surface S. Blow down (-1)-curves until reaching a minimal model $S_{\min}$. KKPY's blowup chemical formula for a point blowup is

$$\mathsf{CF}(\mathrm{Bl}_p\, S) = \mathsf{CF}(S) + \mathsf{CF}(p).$$

A point atom has rank one, whereas α_X has rank twelve. Therefore α_X already occurs on $S_{\min}$.

If $K_{S_{\min}}$ is nef, KKPY's nef-canonical lemma says that $S_{\min}$ has a single atom whose underlying Hod-representation is all of $H^\bullet(S_{\min})$. Its even part has dimension at least three: H^0, an ample class in H^2, and H^4 are linearly independent. This contradicts the even rank 2 of α_X.

If $K_{S_{\min}}$ is not nef, the classification of minimal smooth complex projective surfaces gives either $\mathbf{P}^2$ or a geometrically ruled surface $\mathbf{P}_C(V)$. By KKPY's projective-bundle formula,

$$\mathsf{CF}(\mathbf{P}^2) = 3\mathsf{CF}(\mathrm{pt}), \qquad \mathsf{CF}(\mathbf{P}_C(V)) = 2\mathsf{CF}(C).$$

Thus every atom occurring there is a point or curve atom. Proposition 4.16 excludes α_X in both cases. This proves (4.52). $\square$

4.6 Proof of the one-stabilization theorem

We can now finish without transporting framed monodromy through a birational factorization.

Proof of Theorem 1.1. Since

$$X \times \mathbf{P}^1 = \mathbf{P}_X(\mathcal{O}_X^{\oplus 2}),$$

KKPY's ordinary projective-bundle chemical formula gives

$$\mathsf{CF}(X \times \mathbf{P}^1) = 2\mathsf{CF}(X). \tag{4.53}$$

Hence the Hodge atom α_X occurs with positive multiplicity in the atomic composition of the smooth projective fourfold $X \times \mathbf{P}^1$.

KKPY's ordinary Hodge-atom non-rationality criterion [22, Proposition 5.30] states that if a smooth projective d-fold contains an atom not realizable by any smooth projective variety of dimension at most $d-2$, then the d-fold is not rational. Here $d = 4$, and Proposition 4.17 gives

$$\alpha_X \notin \mathsf{HAtoms}_{\dim \le 2}.$$

Therefore $X \times \mathbf{P}^1$ is irrational. $\square$

Remark 4.18 (Scope of the atom obstruction). The argument is specific to one stabilization. For $X \times \mathbf{P}^m$ with $m \ge 2$, the ordinary atom criterion permits representatives in dimension at least three, and the cubic atom already has the threefold X as such a representative. Stronger stabilization statements therefore require a finer decoration or a different invariant.

5 Framed formal monodromy: a conditional refinement

Section 4 proved the one-stabilization theorem using one ordinary Hodge atom. We now ask for a stronger statement about the entire numerical small even quantum connection: whether its primitive-sixth framed formal-monodromy multiplicity satisfies direct operation formulas under projective bundles and blowups. The actual Iritani and Iritani–Koto comparison isomorphisms preserve the original loop framing after coefficient-field extension. The string equation and divisor equation also remove, exactly, the scalar degree-zero term and any fixed degree-two term whose Novikov character descends to the coefficient ring. What remains is narrower: a residual reconstruction-tail displacement, and a logically separate specialization step in the divisor-tagging argument. We isolate these as Hypotheses 5.7R and 5.7T below.

The results of this section are therefore logically separate from Theorem 1.1. Definition 5.1, the cubic computation, and the product-with-projective-space formula below are unconditional. The projective-bundle and blowup operation formulas use Hypothesis 5.7R. We prove specialized low-dimensional vanishing directly, without Hypothesis 5.7T, for nef-canonical targets, for $\mathbf{P}^1$ and $\mathbf{P}^2$, for every rational geometrically ruled surface—for F_1 on specializations that are graded-monomial in the sense of Definition 5.10, as every center specialization is—and, assuming only Hypothesis 5.7R, for ruled surfaces over curves of positive genus. In the present proof Hypothesis 5.7T is used only for surface centers that are neither minimal nor geometrically ruled. Consequently the resulting birational invariance of ν_6, and the identification $\nu_6(V) = 2$ for genus-eight Fano threefolds, remain conditional on Hypothesis 5.7R and on Hypothesis 5.7T for those centers.

Proposition 4.11 explains how the two quantum invariants in this paper are related on the rank-two block that matters for the cubic. The atomic invariant $\delta^\sharp$ is the square of the difference of the residue representatives selected by the canonical elementary modification, and therefore forgets a common scalar shift. The framed invariant ν_6 is finer: it tests the absolute cyclotomic placement of the formal exponent classes, together with the original-loop descent data when ramification is present. For the cubic zero block, Section 4 already finds the representatives $-1/6$ and $-5/6$; the calculation below recovers those same exponents directly from formal solutions and records that their classes give the two primitive sixth roots. Thus Sections 4 and 5 use two levels of the same formal data rather than unrelated numerical obstructions.

Let T be a smooth projective variety. Write

$$\Lambda_T = \mathbf{C}[[\mathrm{NE}^{\mathbf{Z}}_{\mathrm{num}}(T)]]$$

for the completed monoid ring of the image of effective homology classes in the numerical curve lattice

$$N_1(T) = \mathrm{im}(H_2(T, \mathbf{Z}) \longrightarrow \mathrm{Hom}(\mathrm{NS}(T)/\mathrm{tors}, \mathbf{Z}))\,.$$

Thus $N_1(T)$ is torsion-free and is separated by integral divisors. The completed monoid ring Λ_T is a domain. Indeed, the effective numerical monoid embeds in the torsion-free lattice $N_1(T)$, hence is cancellative and its ordinary monoid algebra is a domain. An ample degree is positive and proper on the effective numerical monoid, so every nonzero completed series has a lowest nonzero homogeneous part, with only finitely many monomials in that degree. The lowest homogeneous part of a product is the product of the two lowest homogeneous parts and is nonzero in the monoid algebra. Therefore $\mathrm{Frac}(\Lambda_T)$ is defined. Let $\mathcal{K}_T$ be an algebraic closure of this fraction field. The loop coordinate z remains independent: in particular, $\mathcal{K}_T$ contains no chosen root of z. We first isolate the cyclotomic part of formal monodromy without choosing values for the exponentials of arbitrary elements of $\mathcal{K}_T$.

Let $K \supset \mathbf{C}$ be an algebraically closed field of characteristic zero. Choose a complement $K = \mathbf{C} \oplus V$ of $\mathbf{Q}$-vector spaces, let $K[V]$ be the group algebra of the additive group of V, and let Ω_V be an algebraic closure of its fraction field. Thus $K \subset \Omega_V$. The map

$$\mathrm{Exp}_V(c + v) = e^{2\pi i c}[v] \colon (K, +) \longrightarrow \Omega_V^\times \tag{5.0}$$

is a group homomorphism. Since V is torsion-free, $K[V]$ is a domain, and its distinct monomials are linearly independent. Consequently

$$\ker(\mathrm{Exp}_V) = \mathbf{Z}, \qquad \mathrm{Exp}_V(a) \text{ is a root of unity} \iff a \in \mathbf{Q}. \tag{5.0a}$$

Apply Levelt–Turrittin after a ramification $z = w^e$. In the formal decomposition, write the residue on a regular-singular block as $R = R_s + R_n$, with commuting semisimple and nilpotent parts. In the universal formal solution algebra, define a full turn of w on that block by

$$M_{\mathrm{RS},V} = \mathrm{Exp}_V(R_s)\exp(2\pi i R_n),$$

where Exp_V is applied to the eigenvalues of R_s. The lift of one turn of the original z-disc acts on the formal symbols by $w \mapsto e^{2\pi i/e}w$, and on the blocks by their descent intertwiners. Its e-th power is $M_{\mathrm{RS},V}$ on every returned regular-singular block. An integral change of residue representative multiplies the formal solution by an integral power of w, so it conjugates this lifted action rather than changing it. Since the coefficient and descent matrices lie in $K \subset \Omega_V$, this defines the framed operator $M_{\mathrm{f},V}$ over Ω_V. Ramified exponential factors are thereby permuted rather than made single-valued by changing the frame.

Choice independence. For every root of unity $\zeta \in \mathbf{C}$, its algebraic multiplicity as an eigenvalue of $M_{\mathrm{f},V}$ is independent of the complement V, the Levelt–Turrittin ramification and decomposition, and the algebraic closure Ω_V. It is also unchanged when the chosen algebraic closure of the original coefficient field is replaced by a common algebraically closed overfield. When $K = \mathbf{C}$, it is the usual multiplicity in the analytic formal monodromy on the original z-disc. We use here the functoriality and uniqueness of the ramified formal decomposition. That statement holds over any algebraically closed coefficient field of characteristic zero, which is the algebraic form we need, and in that generality it is the formal classification of differential modules over $K(\!(z)\!)$ [27, Chapter 3]. Sabbah gives the complex-analytic account, where the same decomposition appears alongside the Riemann–Hilbert correspondence [29, Lecture 5, Section 5.c, proofs of Theorem 5.8 and Proposition 5.10]. Consider one orbit of Levelt–Turrittin blocks under a turn of the original disc, let d be its length, and write $A_i : E_i \longrightarrow E_{i+1}$ for its successive deck/descent maps, with indices modulo d. Put $U = A_{d-1}\cdots A_0$ on E_0. A cyclic-block determinant gives

$$\chi_{\bigoplus_i E_i}(X) = \det(X^d I - U). \tag{5.0b}$$

Put $r = e/d$. Equivariance of the descent datum gives the intrinsic return relation

$$U^r = M_{\mathrm{RS},V}. \tag{5.0c}$$

Thus, if μ is an eigenvalue of U, then $\mu^r = \mathrm{Exp}_V(\rho)$ for a residue exponent ρ, with algebraic multiplicities read on the corresponding generalized eigenspaces. If a root of unity ζ occurs in (5.0b), then $\mu = \zeta^d$ and $\mathrm{Exp}_V(\rho) = \zeta^e$; hence $\rho \in \mathbf{Q}$ by (5.0a). Conversely, if ρ is rational, then μ^r is torsion, so every corresponding μ, and hence every root of $X^d - \mu$, is torsion. On these rational residue classes Exp_V is the usual complex exponential; the finite descent operator is the same intrinsic operator. The exact roots of unity and their algebraic multiplicities are therefore those of the ordinary complex construction. Nonrational residue classes contribute no root of unity. This criterion does not depend on V and does not require a fractional residue factor to commute with the descent action.

For completeness, compare two Levelt–Turrittin choices after pulling both to a common ramification $z = \widetilde{w}^E$. Equivariant uniqueness of the formal decomposition identifies their total horizontal solution spaces and carries the original-z-turn deck/descent operator for one choice to the scalar extension of the other. Thus the product of all orbit polynomials (5.0b), rather than any individually labelled return map, is invariant under refinement and re-decomposition. The same argument after a common algebraically closed coefficient extension preserves the intrinsic multiset of residue classes and finite descent characters. The relation (5.0c) is itself invariant under integral changes of residue representative. Hence the cyclotomic test above and its algebraic multiplicities depend only on the original framed module, not on the coefficient closure or complement. Finally, for $K = \mathbf{C}$ one has $V = 0$ and $\mathrm{Exp}_V(a) = e^{2\pi i a}$, which gives the usual analytic formal monodromy.

Definition 5.1. Apply the preceding construction to the small even quantum connection over $\mathcal{K}_T((z))$, and put

$$\nu_6(T) = \operatorname{mult}_{e^{\pi i/3}} M_{\mathrm{f},V}(T) + \operatorname{mult}_{e^{-\pi i/3}} M_{\mathrm{f},V}(T).$$

The multiplicities are algebraic multiplicities in the characteristic polynomial. The choice-independence argument above makes their sum independent of all the auxiliary choices.

The frame matters. Adjoining roots of Novikov monomials is an algebraic coefficient extension and is harmless by choice independence. Adjoining a root of z, by contrast, would replace the original loop and could erase the deck part of the operator.

Lemma 5.1A (Frame transport over a coefficient field)**.** *Let $K \supset \mathbf{C}$ be an algebraically closed field of characteristic zero, and let D, D' be differential modules over $K(\!(z)\!)$. Suppose $G \in \mathrm{GL}_n(K(\!(z)\!))$ is a gauge isomorphism from D to D'. Then the framed operators $M_{\mathrm{f},V}(D)$ and $M_{\mathrm{f},V}(D')$, defined using the original z-disc turn, are conjugate after passage to a common coefficient extension. In particular the algebraic multiplicity of every root of unity is the same for D and D'.*

Proof. After a common Levelt–Turrittin ramification, choose a formal solution matrix Y for D in the formal solution algebra used in Definition 5.1, and let σ denote one turn of the original z-disc. Since G has entries in $K(\!(z)\!)$, $\sigma(G) = G$. Hence

$$\sigma(GY) = G\sigma(Y).$$

Thus GY is a solution frame for D' in which the original-disc turn has the same matrix as in the frame Y. Comparing with any other solution frame for D' conjugates that matrix by a constant matrix. Therefore the framed operators are conjugate and have the same characteristic polynomial. □

Lemma 5.2 (Exact string and fixed-divisor shifts)**.** *Let a pullback of the even quantum connection of a smooth projective variety T be defined over a coefficient field containing its Novikov coefficients.*

(i) Replacing a bulk point τ by $\tau + c\mathbf{1}$ changes the z-connection only by the scalar irregular twist $e^{-c/z}$. Hence it does not change the framed formal-monodromy eigenvalues on the original z-disc, for any coefficient c for which the shifted connection is defined.

(ii) Let $D \in H^2(T) \otimes K$ be a bulk shift such that $d \mapsto e^{\langle D,d\rangle}$ defines a multiplicative character on the numerical Novikov image in the coefficient field. Then the connection at $\tau + D$ is obtained from that at τ by the coefficient substitution

$$Q^d \longmapsto e^{\langle D,d\rangle} Q^d.$$

This substitution fixes $\mathbf{C}$, so it preserves the algebraic multiplicities of every root of unity in framed formal monodromy.

Proof. Write the quantum connection in the convention of [19, (2.2)–(2.3) and Remark 2.3]. The String Equation implies that the quantum product is independent of the H^0 coordinate, while the Euler field acquires $c\mathbf{1}$. Thus the horizontal equation $z^2\partial_z Y = (U - z\mu)Y$ acquires the scalar term cI, and multiplication of a solution by $e^{-c/z}$ gives the shifted equation. This exponential is single-valued under a full turn of the original z-disc and changes no regular exponent or descent datum.

For a degree-two shift, the Divisor Equation says that the quantum product depends on the divisor coordinate only through the combinations $Q^d e^{\langle D,d\rangle}$; the degree-two coordinate does not occur in the Euler field. Therefore the full z-connection is the coefficient base change displayed above. The induced embedding of coefficient fields fixes the complex roots of unity, and choice independence in Definition 5.1 gives the last assertion. □

Remark 5.3 (Why a residual hypothesis remains). Flatness does give a normalized gauge for a genuinely formal bulk germ. Writing the formal bulk variables collectively as η and solving

$$\partial_{\eta_i} G = -z^{-1}(\phi_i \star_\eta) G, \qquad G|_{\eta=0} = I,$$

recursively, the coefficient of total bulk degree m has no power of z below $-m$. Thus every fixed bulk degree uses only integral powers of the original loop coordinate and is harmless for framing. After substituting a reconstruction coordinate containing Novikov coefficients, however, the sum over all bulk degrees need not have a uniform lower Laurent bound in z. We therefore do not promote this degreewise gauge to an element of a Laurent series field and do not use it to identify intrinsic framed monodromy at a reconstruction value. Hypotheses 5.7R and 5.7T below isolate exactly the two places where such an identification is still required. The rank-two theorem below avoids this gauge altogether on the formal germ where it applies.

Lemma 5.4 (Numerical Novikov base change)**.** *The quotient from the effective homology monoid to its image in $N_1(T)$ extends continuously to completed monoid rings, and the quantum connection and the two comparison theorems base-change along it.*

Proof. Filter both monoids by intersection with an ample divisor. Effective homology classes below a fixed cutoff form a finite set, as follows from the finite-type Chow varieties of cycles of bounded degree. The quotient is therefore defined on every truncated monoid ring and passes to the inverse limit. Explicitly, the coefficient of $Q^{\bar{d}}$ in the image of $\sum_d a_d Q^d$ is

$$\sum_{d \mapsto \bar{d}} a_d.$$

The sum is finite at every cutoff, and additivity of the quotient makes this coefficient-wise pushforward a unital homomorphism for completed convolution. Thus the quantum products descend by grouping Gromov–Witten coefficients with the same numerical class. Pushforward of curves, pairing with $c_1(N_{C/T})$, and the derivations $Q\partial_Q$ are constant on these fibres. At each cutoff, apply the resulting ring homomorphism to the comparison map and to its stated inverse. Scalar extension preserves their two inverse identities. Grouping coefficients identifies the scalar-extended quantum module with the numerical quantum module, and the preceding fiberwise constancy identifies its connection operators and divisor substitutions. Passing through the compatible cutoffs proves the asserted completed base change. □

Proposition 5.5 (Formal-germ rigidity of an isolated rank-two packet)**.** *Let K be an algebraically closed coefficient field of characteristic zero and let the even quantum connection be pulled back to a formal smooth bulk germ $B = K[[t_1, \ldots, t_m]]$. Suppose that at the closed point the Euler operator has an eigenvalue whose generalized eigenspace has rank two, its nilpotent part is a single nonzero Jordan block, and every other Euler eigenvalue differs from it by a unit of K. Then the corresponding rank-two spectral cluster remains a single nonzero Jordan block on the formal germ. After scalar centering and the canonical elementary modification of Section 4, its residue R has constant characteristic polynomial on B.*

Consequently the algebraic multiplicities of the primitive sixth roots contributed by this block are constant on the formal germ.

Proof. The unit separation from the other Euler eigenvalues gives a formal spectral projector onto the rank-two cluster. Put $U_0 = \lambda I + N$ on this cluster, with $\operatorname{tr} N = 0$, and let C_a denote quantum multiplication by the bulk tangent vector ∂_{t_a}. Flatness gives the coefficient identities

$$[U, C_a] = 0, \qquad \partial_{t_a} U = C_a + [C_a, \mu].$$

Let P be the formal spectral projector onto this cluster and use the induced connection D_a on $P(H)$. For an endomorphism preserving the cluster, D_a is the projected derivative, so

$$D_a U_0 = P(\partial_{t_a} U)P.$$

Because $[U, C_a] = 0$, the operator C_a commutes with P. Compressing the second flatness identity therefore gives

$$D_a U_0 = C_{a,0} + [C_{a,0}, \mu_0], \qquad C_{a,0} = PC_aP, \quad \mu_0 = P\mu P.$$

Because the reduction of U_0 at the closed point is nonscalar, U_0 is a regular 2×2 matrix over the local ring B. After choosing the closed-point Jordan frame, one off-diagonal entry of N is a unit; solving $[N, X] = 0$ directly in that frame gives the commutant $BI \oplus BN$. Hence

$$C_{a,0} = p_a I + q_a N.$$

Taking traces in the block evolution gives $p_a = \partial_{t_a} \lambda$ and therefore

$$D_a N = q_a N + q_a [N, \mu_0].$$

Set $d = \det N$. Since determinant is invariant under change of frame, its ordinary derivative can be computed with the covariant derivative D_a. For a traceless 2×2 matrix, $\operatorname{adj}(N) = -N$ and $N^2 = -dI$. Cyclicity of trace then gives

$$\partial_{t_a} d = \operatorname{tr}(\operatorname{adj}(N) D_a N) = 2q_a d.$$

Since d vanishes at the closed point, a lowest-homogeneous-term argument in $K[[t_1, \dots, t_m]]$ forces $d = 0$. Thus $N^2 = 0$ everywhere. A matrix entry of N is a unit at the closed point, so N remains nonzero and its image is a line subbundle.

The proof of Proposition 4.10 is purely algebraic: it uses only flatness, Frobenius self-adjointness, anti-self-adjointness of the grading operator, and the nondegenerate Poincaré pairing on the separated spectral factor. Those identities hold coefficientwise over the present formal germ, so the same elementary modification is regular in every base direction and its residue satisfies

$$D_a R = [G_a, R].$$

Hence $\operatorname{tr} R$ and $\det R$, and therefore the characteristic polynomial of R, are constant. By Proposition 4.11, the residue eigenvalues are representatives of the formal exponent classes; the integral shear changes them only by integers. The eigenvalues of the regular formal monodromy are therefore constant, including in resonant cases where only the unipotent part could vary. This proves the cyclotomic multiplicity statement. □

Remark 5.6 (Formal-germ scope). Proposition 5.5 is deliberately a theorem over a formal bulk germ with a fixed coefficient field. It does not identify an Iritani or Iritani–Koto reconstruction value with a point of that germ. The comparison theorems are formulated over graded Laurent completions in Novikov variables, whereas the proposition deforms genuinely formal bulk variables over K. Substituting a Laurent-Novikov reconstruction expression for those formal variables is an additional operation, and no theorem legitimizing that substitution for framed monodromy is assumed here. Thus the proposition proves the local rank-two mechanism behind Hypothesis 5.7R; it is not a proof of that hypothesis at a reconstruction value.

Hypothesis 5.7R (Reconstruction-tail invariance)**.** *Consider only the residual reconstruction displacements that occur in the blowup and projective-bundle decompositions used in Proposition 5.7: the $O(u) + s_j$ tails on the target summands and the corresponding residual ambient shift, after the scalar H^0 term and every fixed H^2 term whose Novikov character descends to the relevant coefficient ring have been removed by Lemma 5.2. After the coefficient specialization displayed in the construction, replacing zero bulk by one of these residual reconstruction displacements does not change the algebraic multiplicities of $e^{\pi i/3}$ and $e^{-\pi i/3}$ in framed formal monodromy on the original z-disc.*

No invariance under an arbitrary bulk displacement is asserted.

Proposition 5.7 (Framed operation formulas, conditional)**.** *Assume Hypothesis 5.7R. Let $V \to T$ be a vector bundle of rank $r \geq 2$. Then*

$$\nu_6(\mathbf{P}_T(V)) = r\nu_6(T). \tag{5.2}$$

Let $\widetilde{T} = \mathrm{Bl}_C T$, where $C \subset T$ is smooth of codimension $c \geq 2$. In Iritani's comparison field one has

$$\nu_6(\widetilde{T}) = \nu_6(T) + \sum_{j=0}^{c-2} \nu_6(C; \chi_j), \tag{5.3}$$

where χ_j is the numerical Novikov specialization of the j-th center summand. The endpoint terms in (5.3) are the intrinsic small invariants; the center terms are computed after the displayed specializations.

Proof. We first treat the blowup. Iritani's Theorem 5.18 gives a formal coordinate isomorphism and a quantum-D-module decomposition

$$\mathrm{QDM}(\widetilde{T}) \cong \mathrm{QDM}(T) \oplus \mathrm{QDM}(C)^{\oplus(c-1)}$$

over a Laurent extension in the exceptional Novikov variable [19, Theorem 5.18]. The comparison map and its inverse use only integral powers of the original loop coordinate z. After passing the coefficient ring to an algebraically closed Laurent coefficient field, Lemma 5.1A therefore transports framed formal monodromy across the comparison without ramifying the z-disc.

For a center summand put $u = q^{-1/(c-1)}$. Iritani's reconstruction coordinates have the initial form

$$\varsigma_j = \varsigma_j^\circ + s_j, \qquad \varsigma_j^\circ = -(c-1)\lambda_j + h_{C,j} + O(u), \tag{5.4}$$

with $\lambda_j = \zeta_j u^{-1}$ [19, Section 5.8.2 and (5.27)–(5.30)]. The associated numerical Novikov specialization is

$$\chi_j(Q_C^d) = Q^{i_* d} u^{\rho_C \cdot d}. \tag{5.4a}$$

The unnormalized connection is naturally read after the Laurent localization containing u^{-1}; in particular we make no claim that the term λ_j belongs to a nonlocalized center ring. Lemma 5.2 removes the scalar H^0 term $-(c-1)\lambda_j$ and the fixed divisor $h_{C,j}$ exactly. The latter character descends through (5.4a) because $h_{C,j}$ is a scalar multiple of ρ_C. The remaining $O(u) + s_j$ reconstruction tail is the center instance covered by Hypothesis 5.7R. The normalization of the logarithmic coordinate is fixed in [19, Section 5.8.1]; a residual constant logarithmic character would again be a fixed divisor shift and hence is already covered by Lemma 5.2.

The ambient summand is analogous. Its reconstruction coordinate is $\tau = \tau^\circ + t$ with $\tau^\circ = O(q^{-1})$ in the Laurent expansion of Theorem 5.18. Hypothesis 5.7R is invoked only for this residual ambient reconstruction tail. Thus, after the exact low-degree normalizations, the ambient contribution has the intrinsic multiplicity of T, while the j-th center contribution has the intrinsic multiplicity of the small connection of C after the specialization χ_j. Direct-sum additivity gives (5.3).

The coefficient extensions do not alter the endpoint invariants. On numerical curve lattices the blowup map

$$\bar{\beta} \longmapsto (\phi_* \bar{\beta}, -E \cdot \bar{\beta})$$

is injective because $N^1(\widetilde{T}) = \phi^* N^1(T) \oplus \mathbf{Z}[E]$. Hence the numerical Novikov field of $\widetilde{T}$ embeds in the comparison field. The center map (5.4a) need not be injective, which is why its contribution is denoted $\nu_6(C; \chi_j)$ rather than $\nu_6(C)$.

For a projective bundle, Iritani–Koto's Theorem 5.1 gives the corresponding decomposition into r copies of the quantum D-module of T over a Laurent coefficient extension [21, Theorem 5.1]. Again the comparison and its inverse use only integral powers of the same loop coordinate, so Lemma 5.1A applies. With $u = q^{-1/r}$ their reconstruction coordinates satisfy

$$\varsigma_j = \varsigma_j^\circ + s_j, \qquad \varsigma_j^\circ = r\lambda_j - \frac{2\pi \mathrm{i} j}{r} c_1(V) + O(u)$$

[21, Theorem 5.1(4), Proposition 5.6]. Lemma 5.2 removes the scalar term and the fixed divisor term. The initial-coordinate normalization is fixed by [21, (5.11)–(5.12)]. Hypothesis 5.7R is then used only for the residual $O(u) + s_j$ reconstruction tail. The base Novikov map is injective [21, (1.1) and Remark 5.2]; hence every summand contributes the intrinsic value $\nu_6(T)$, and direct-sum additivity gives (5.2).

The projective-bundle theorem is stated after a global-generation normalization. Tensoring V by a sufficiently negative line bundle does not change $\mathbf{P}_T(V)$, and the intrinsic Novikov embedding is unchanged by this twist [21, Remark 1.2 and Remark 5.2]. Finally, both comparison decompositions preserve cohomological parity on the even restriction used here. Iritani records this explicitly for blowups [20, Section 2]; for projective bundles it follows directly from the pullback, pushforward, Fourier-transform, and characteristic-class operations in Theorem 5.1 after odd bulk variables are set to zero. The adjoined Novikov roots are coefficient variables, not roots of z. □

The center map in (5.3) is not necessarily injective [19, (5.15)]. Thus generic vanishing for C does not by itself imply that its specialized term vanishes. The next lemma supplies the missing implication.

Definition 5.8. A monomial map $\chi : \Lambda_T \to A$ is *strictly Novikov-admissible* if A is a complete separated valued $\mathbf{C}$-domain with domain associated graded, every monomial has nonzero image, and

$$v(\chi(Q^d)) = \ell(d), \tag{5.5}$$

where ℓ is positive and proper on the effective numerical monoid.

We write $\nu_6(T; \chi)$ for the framed small-even invariant after scalar extension along such a map.

The center maps in (5.3) are strictly admissible. Give the exceptional Laurent variable valuation zero and use an ample divisor H on T. Then

$$v_H\Big(Q^{i_*d}q^{-\rho_C\cdot d/(c-1)}\Big) = (H|_C)\cdot d.$$

The restriction $H|_C$ is ample, which gives positivity and properness; the associated graded Laurent monoid ring is a domain.

Proposition 5.9 (Direct specialized low-dimensional vanishing)**.** *Let $\chi : \Lambda_T \to A$ be strictly Novikov-admissible.*

(i) If the canonical divisor K_T is nef, then

$$\nu_6(T;\chi) = 0$$

unconditionally.

(ii) If $T = \mathbf{P}^1$ or $T = \mathbf{P}^2$, then

$$\nu_6(T;\chi) = 0$$

unconditionally.

(iii) Let $T = \mathbf{P}_C(V)$ be a geometrically ruled surface with $g(C) \geq 1$. Assuming Hypothesis 5.7R,

$$\nu_6(T;\chi) = 0.$$

Proof. For (i), work after scalar extension to an algebraic closure of $\operatorname{Frac} A$. At small bulk write

$$\nabla_{z\partial_z} = z\partial_z - z^{-1}U + \mathrm{Gr}, \qquad U = \sum_\beta \chi(Q^\beta)U_\beta,$$

where U_β is the coefficient of Euler multiplication in numerical class β. Put

$$g = \mathrm{Gr} + \frac{1}{2}E,$$

with E equal to the identity on cohomological degrees of parity opposite to $\dim T$ and zero on the other parity. The operator g has integral eigenvalues. The genus-zero dimension constraint gives

$$U_\beta : H^i(T) \longrightarrow H^{i+2-2c_1(T)\cdot\beta}(T),$$

so, because quantum multiplication preserves parity, U_β raises g-weight by

$$m_\beta = 1 - c_1(T)\cdot\beta.$$

If K_T is nef, then $m_\beta \geq 1$ for every effective β. Make the half-parity correction explicitly by replacing the z-connection with

$$\nabla^{\mathrm{par}}_{z\partial_z} = \nabla_{z\partial_z} + \frac{1}{2}E = z\partial_z - z^{-1}U + g.$$

With the gauge convention $G\nabla G^{-1}$, take $G = z^g$. Since $[g, U_\beta] = m_\beta U_\beta$, one has

$$z^g U_\beta z^{-g} = z^{m_\beta}U_\beta, \qquad z^g(z\partial_z)z^{-g} = z\partial_z - g,$$

and therefore

$$z^g\nabla^{\mathrm{par}}_{z\partial_z}z^{-g} = z\partial_z - \sum_\beta \chi(Q^\beta)z^{m_\beta-1}U_\beta.$$

Thus the specialized connection is regular singular. This conclusion is unchanged when the noninjective map χ groups several numerical classes: every grouped summand still has nonnegative z-order. The residue consists only of the terms with $m_\beta = 1$, equivalently $c_1(T) \cdot \beta = 0$. Each such operator raises the finite g-filtration by one, so their sum is nilpotent. The parity-corrected regular monodromy is therefore unipotent. Undoing the half-parity correction permits only eigenvalues 1 and -1, because the transformed residue preserves parity. Hence no primitive sixth root occurs. Strict Novikov-admissibility is used here only to ensure that the specialized small connection is a well-defined coefficientwise specialization; injectivity is not used.

For (ii), let ℓ be the line class and put $a = \chi(Q^\ell)$. Strict admissibility gives $a \neq 0$. Over an algebraic closure of $\operatorname{Frac} A$, the specialized small quantum relation for $\mathbf{P}^m$, $m = 1, 2$, is

$$H^{m+1} = a.$$

Euler multiplication by $(m+1)H$ therefore has the $m+1$ distinct eigenvalues

$$(m+1)\zeta a^{1/(m+1)}, \qquad \zeta^{m+1} = 1.$$

Every spectral block has rank one, so Lemma 5.20 gives trivial framed regular monodromy on every block.

For (iii), let f be the fibre class. Every map from a connected nodal genus-zero curve to C is constant: it is constant on every irreducible rational component, and connectedness forces the same image point on all components. Hence every nonzero curve class contributing to the genus-zero quantum product of T is a positive multiple of f. The small quantum differential module is therefore obtained by scalar extension from a one-variable module over $\mathbf{C}[[Q^f]]$. The restriction of χ to the fibre ray is injective, because

$$v(\chi(Q^{nf})) = n\ell(f), \qquad \ell(f) > 0,$$

so distinct powers have distinct valuation. On the other hand, Hypothesis 5.7R and Proposition 5.7 give the intrinsic projective-bundle identity

$$\nu_6(T) = 2\nu_6(C) = 0,$$

where the last equality follows by applying (i) to the identity Novikov specialization of C, since K_C is nef for $g(C) \geq 1$. The intrinsic small module and the χ-specialized module are both scalar extensions of this same one-variable differential module. More explicitly, let

$$K_f = \operatorname{Frac}(\mathbf{C}[[Q^f]]), \qquad K_{f,\chi} = \operatorname{Frac}(\chi(\mathbf{C}[[Q^f]])) \subset \operatorname{Frac}(A).$$

The injective restriction of χ identifies K_f with $K_{f,\chi}$. Via this identification, choose a common algebraically closed overfield Ω containing K_f and $\operatorname{Frac}(A)$. After scalar extension to Ω, the intrinsic one-variable module and its χ-specialization become literally the same differential module. Definition 5.1 and its choice-independence under common coefficient-field extension therefore give identical root-of-unity multiplicities. The specialized value also vanishes. $\square$

The remaining centers are the rational geometrically ruled surfaces, that is the Hirzebruch surfaces $F_a = \mathbf{P}_{\mathbf{P}^1}(\mathcal{O} \oplus \mathcal{O}(a))$; these are the minimal surfaces not yet covered, together with F_1, which is geometrically ruled without being minimal. For those the projective-bundle comparison is available only intrinsically, so we compute the specialized Euler spectrum instead. The quadric surface $F_0 = \mathbf{P}^1 \times \mathbf{P}^1$ is handled separately by the Gromov–Witten product formula, because χ may send its two ruling classes to the same value and the quartic below then degenerates. For $a \geq 1$ two facts carry the computation: one deformation carries the genus-zero theory of F_a to that of F_0 or F_1, whose quantum relations serve as the base cases, under an explicit change of Novikov variables, and the resulting rank-four Euler quartic is separable away from one explicit locus that no center specialization meets.

One case, the surface F_1, needs a mild extra condition on χ, used only to rule out a single algebraic coincidence in the associated graded ring. The word *monomial* in Definition 5.8 refers to the source, a map defined on Novikov monomials; the condition below is about the target, so we call it graded-monomial.

Definition 5.10. A strictly Novikov-admissible $\chi : \Lambda_T \to A$ is *graded-monomial* if the associated graded ring $\operatorname{gr} A$ admits a $\mathbf{C}$-basis B containing the leading term $\operatorname{gr} \chi(Q^d)$ of $\chi(Q^d)$ for every effective class d.

Lemma 5.11 (Center specializations are graded-monomial)**.** *The specializations* (5.4a) *are graded-monomial.*

Proof. The associated graded of the target ring A of (5.4a) is the Laurent monoid ring generated by the numerical Novikov monomials of T and the exceptional variable, and its monomials form a $\mathbf{C}$-basis. Each $\chi_j(Q_C^d) = Q^{i_* d} u^{\rho_C \cdot d}$ is one of them, with coefficient 1. ☐

Lemma 5.12 (Specialized Euler spectrum of a Hirzebruch surface)**.** *Let $T = F_a = \mathbf{P}_{\mathbf{P}^1}(\mathcal{O} \oplus \mathcal{O}(a))$ with $a \geq 0$, let f and s be the class of a fibre and the class of a section of self-intersection $-a$ in $N_1(T)$, and put $k = \lfloor a/2 \rfloor$. For $a = 0$ either ruling may be taken as the fibre; the quartic below is symmetric in u and w. Let $\chi : \Lambda_T \to A$ be strictly Novikov-admissible and set*

$$u = \chi(Q^f), \qquad w = \chi(Q^{s+kf}).$$

After scalar extension to an algebraic closure of $\operatorname{Frac} A$*, the characteristic polynomial of Euler multiplication on $H^0(T) \oplus H^2(T) \oplus H^4(T)$ is*

$$P_a(X) = \begin{cases} X^4 - 8(u+w)X^2 + 16(u-w)^2, & a \text{ even}, \\ X^4 + wX^3 - 8uX^2 - 36uwX + 16u^2 - 27uw^2, & a \text{ odd}, \end{cases} \tag{5.5a}$$

and its discriminant is

$$\operatorname{disc} P_a = \begin{cases} 2^{24}\, u^2 w^2 (u-w)^2, & a \text{ even}, \\ -u^2 w^2 \left(256u + 27w^2\right)^3, & a \text{ odd}. \end{cases} \tag{5.5b}$$

Proof. Write F and S for the divisor classes corresponding to f and s, the class of a fibre and the class of the section of self-intersection $-a$, so that $F^2 = 0$, $S \cdot F = 1$, $S^2 = -a$ and $c_1(T) = 2S + (a+2)F$. Put $S^\flat = S + kF$, of self-intersection $-a + 2k \in \{0, -1\}$; then

$$c_1(T) = \begin{cases} 2S^\flat + 2F, & a = 2k, \\ 2S^\flat + 3F, & a = 2k+1. \end{cases}$$

A single deformation to $a \leq 1$. Let $a \geq 2$. One family is used rather than k successive ones, which keeps the transport explicit. Since $k \geq 1$ and $a - k \geq 1$, a general pair of sections of $\mathcal{O}(k)$ and $\mathcal{O}(a-k)$ has no common zero, so it defines an injection $\mathcal{O} \to \mathcal{O}(k) \oplus \mathcal{O}(a-k)$ with locally free quotient, necessarily $\mathcal{O}(a)$ by degree. Let $\varepsilon \in \operatorname{Ext}^1(\mathcal{O}(a), \mathcal{O})$ be the class of the resulting extension and consider the extensions E_t with class $t\varepsilon$, $t \in \mathbf{A}^1$, that is the pullback of the universal extension along $t \mapsto t\varepsilon$. These form a vector bundle on $\mathbf{P}^1 \times \mathbf{A}^1$. Scaling a nonzero extension class by a unit leaves the middle term unchanged, so $E_t \cong E_1$

for every $t \neq 0$. The extension is nonsplit, since $\mathcal{O}(k) \oplus \mathcal{O}(a-k)$ is not $\mathcal{O} \oplus \mathcal{O}(a)$; and its projectivization is a smooth projective family with

$$\mathbf{P}(E_0) = \mathbf{P}(\mathcal{O} \oplus \mathcal{O}(a)) = F_a, \qquad \mathbf{P}(E_t) = \mathbf{P}(\mathcal{O}(k) \oplus \mathcal{O}(a-k)) \cong F_{a-2k} \quad (t \neq 0),$$

the last isomorphism by twisting with $\mathcal{O}(-k)$. Genus-zero Gromov–Witten invariants are invariants of a deformation class [24, Chapter 7]; a smooth projective family carries a smooth family of Kähler forms, and every symplectic four-manifold is semipositive, so the genus-zero invariants of [24] are defined and deformation-invariant for this family. The invariants therefore agree under the parallel transport ψ of N_1 along the family. The ruling is global over the base, so ψ fixes f; writing $\psi(s) = \alpha s_0 + \beta f$ with s_0 the negative-section class of F_{a-2k}, which for F_0 means the other ruling class, pairing with f gives $\alpha = 1$ and $\psi(s)^2 = -a$ gives $\beta = -k$. Thus

$$\psi(mf + ns) = (m - nk)f + ns_0, \qquad \psi(f) = f, \quad \psi(s + kf) = s_0,$$

and correspondingly $S^\flat \mapsto S_0$, the negative section of $F_{a-2k} = F_{a \bmod 2}$ as a divisor class. For $a \leq 1$ take ψ to be the identity.

The two base cases. For $F_0 = \mathbf{P}^1 \times \mathbf{P}^1$ the Gromov–Witten product formula [5] gives $QH^*(F_0) = QH^*(\mathbf{P}^1) \otimes QH^*(\mathbf{P}^1)$, so

$$S_0 \star S_0 = Q^f, \qquad F \star F = Q^{s_0}.$$

For F_1, which is Fano, the toric quantum Stanley–Reisner presentation of [4, Section 5]—established for Fano toric manifolds by the mirror theorem [14, Theorem 0.1]—gives

$$S_0 \star S_0 + S_0 \star F = Q^f, \qquad F \star F = Q^{s_0} S_0.$$

The same two relations can be read off directly from the genus-zero invariants of F_1, which is how we use them; the toric presentation is recorded as a check. On F_1, $c_1 \cdot (mf + ns_0) = 2m + n$, so a three-point invariant with two divisor insertions and $\beta \neq 0$ is nonzero only for $1 \leq 2m + n \leq 2$; the contributing classes are s_0, with $\langle\ \rangle_{s_0} = 1$ because the negative section is the unique curve in its class, and f, with $\langle \mathrm{pt} \rangle_f = 1$ because one fibre passes through a general point. The class $2s_0$ contributes nothing, since every stable map in it has image the negative section and a general point is not on that curve. The divisor axiom then gives $S_0 \star S_0 = -\mathrm{pt} + Q^{s_0} S_0 + Q^f$, $S_0 \star F = \mathrm{pt} - Q^{s_0} S_0$ and $F \star F = Q^{s_0} S_0$, which is the displayed presentation.

Truncation. The transported theory determines the small quantum product of F_a itself, and only finitely many Novikov monomials survive. By the virtual dimension count, and by the fundamental-class axiom at the lower end, a three-point invariant of F_a with two divisor insertions and $\beta \neq 0$ vanishes unless $1 \leq c_1(T) \cdot \beta \leq 2$; and by deformation invariance it vanishes unless $\psi(\beta)$ is effective on $F_{a \bmod 2}$, that is unless $m \geq nk$ and $n \geq 0$ for $\beta = mf + ns$. Now

$$c_1(T) \cdot \beta = 2m + (2-a)n = \begin{cases} 2(m - nk) + 2n, & a = 2k, \\ 2(m - nk) + n, & a = 2k + 1, \end{cases}$$

so $c_1(T) \cdot \beta \leq 2$ forces $n \leq 1$ for a even and $n \leq 2$ for a odd, and then $m = nk$ except for $\beta = f$. The surviving classes are f and $s + kf$ for a even, and f, $s + kf$ and $2(s + kf)$ for a odd. In particular the product involves no infinite Novikov sum, and the two monomials Q^f and Q^{s+kf} generate every coefficient that occurs. The displayed relations therefore hold in the specialized ring, and the resulting surjection from the presented algebra, which is free of rank four, onto the specialized quantum cohomology, also free of rank four, is an isomorphism.

The quartic. Specializing along χ and substituting $Q^f \mapsto u$ and $Q^{s+kf} \mapsto w$ in the transported presentation, the specialized small quantum cohomology of F_a is $A[S^\flat, F]$ modulo $(S^{\flat 2} - u,\ F^2 - w)$ for a even and modulo $(S^{\flat 2} + S^\flat F - u,\ F^2 - wS^\flat)$ for a odd. In the basis $1, F, S^\flat, S^\flat F$ the matrix of multiplication by $c_1(T)$ is

$$\begin{pmatrix} 0 & 2w & 2u & 0 \\ 2 & 0 & 0 & 2u \\ 2 & 0 & 0 & 2w \\ 0 & 2 & 2 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 0 & 2u & wu \\ 3 & 0 & 0 & 2u \\ 2 & 3w & 0 & 0 \\ 0 & 2 & 1 & -w \end{pmatrix}$$

respectively, and its characteristic polynomial is (5.5a). Evaluating the discriminant of a monic quartic on those coefficients gives (5.5b). □

Lemma 5.13 (Degeneracy dichotomy of the ruled Euler spectrum)**.** *Keep the notation of Lemma 5.12; strict admissibility makes u and w nonzero. Exactly one of the following holds.*

(a) $\operatorname{disc} P_a \neq 0$. *Equivalently $u \neq w$ for a even and $256u + 27w^2 \neq 0$ for a odd. Then every spectral block of Euler multiplication has rank one.*

(b) $\operatorname{disc} P_a = 0$. *Then P_a has exactly one double root and two simple roots, so Euler multiplication has one spectral block of rank two, whose nilpotent part squares to zero, and two blocks of rank one.*

No other block shape occurs. By Lemma 5.14, case (b) does not arise for a center specialization when $a \geq 1$; for F_0 it can, which is why that surface is treated by the product formula in Proposition 5.15.

Proof. By (5.5b) and $u, w \neq 0$, the discriminant vanishes exactly on the stated locus. Off it the quartic is separable, so each of its four roots is simple and each maximal generalized eigenspace of Euler multiplication is a line.

On the locus the quartic factors. For a even, $u = w$ gives $P_a = X^2(X^2 - 16u)$, whose double root is 0 and whose two simple roots are the square roots of $16u$, distinct from 0 and from each other. For a odd, write $w = 16\sigma$; then $256u + 27w^2 = 0$ reads $u = -27\sigma^2$ and

$$P_a = (X + 18\sigma)^2 \left(X^2 - 20\sigma X + 612\sigma^2\right),$$

whose quadratic factor has the two roots $10\sigma \pm 16\varepsilon$ with $\varepsilon^2 = -2\sigma^2$; these are distinct from each other and from -18σ: a coincidence $10\sigma \pm 16\varepsilon = -18\sigma$ gives $4\varepsilon = \mp 7\sigma$, hence $49\sigma^2 = 16\varepsilon^2 = -32\sigma^2$ and $81\sigma^2 = 0$. In both cases the multiplicity of every root is at most two and the repeated root has multiplicity exactly two. A block of rank two has a single eigenvalue λ, hence trace 2λ and determinant λ^2; by Cayley–Hamilton its centered matrix $M - \lambda$ squares to zero. □

Lemma 5.14 (Strictly admissible specializations avoid the degeneracy locus for $a \geq 1$)**.** *Let $T = F_a$ with $a \geq 1$ and let $\chi : \Lambda_T \to A$ be strictly Novikov-admissible, and graded-monomial if $a = 1$. Then* $\operatorname{disc} P_a \neq 0$.

Proof. Write $p = \ell(f)$ and $r = \ell(s)$; both are positive. Multiplicativity of the valuation and (5.5) give $v(u) = p$ and $v(w) = r + kp$.

Let $a = 2k$ with $k \geq 1$. Then $v(w) = r + kp > p = v(u)$, so $u \neq w$.

Let $a = 2k + 1$ with $k \geq 1$. Then $v(w^2) = 2r + 2kp > p = v(u)$, and a vanishing $256u + 27w^2$ would make $256u = -27w^2$. The integers 256 and 27 are units of A, and negation does not change a valuation, so this forces $v(u) = v(w^2)$, a contradiction.

Let $a = 1$, so $k = 0$ and $v(u) = p$, $v(w^2) = 2r$. If $p \neq 2r$ the previous argument applies. If $p = 2r$, work in the degree-p symbol group. Write $\sigma_p : A_{\geq p} \to A_{\geq p}/A_{>p}$ for the symbol map, which is additive. Both $256u$ and $27w^2$ have valuation exactly p, so

$$\sigma_p(256u + 27w^2) = 256\,\sigma_p(u) + 27\,\sigma_p(w^2),$$

and $\sigma_p(u) = \operatorname{gr}\chi(Q^f)$ and $\sigma_p(w^2) = \operatorname{gr}\chi(Q^{2s})$, the second because χ is a monoid map, so $w^2 = \chi(Q^{2s})$. Both are members of the basis B of Definition 5.10, since f and $2s$ are effective. Distinct basis members have no vanishing combination with nonzero coefficients, and equal ones would require $256+27 = 0$, so the symbol is nonzero and therefore $256u+27w^2 \neq 0$. □

Proposition 5.15 (Direct specialized vanishing for rational geometrically ruled centers)**.** *Let $T = F_a$ with $a \geq 0$ and let $\chi : \Lambda_T \to A$ be strictly Novikov-admissible, and graded-monomial if $a = 1$. Then*

$$\nu_6(T; \chi) = 0.$$

Proof. Let $a = 0$, so $T = \mathbf{P}^1 \times \mathbf{P}^1$; this case is the χ-specialized analogue of Proposition 5.21 below. The Gromov–Witten product formula [5] identifies the small quantum differential module of a product with the tensor product of those of its factors over $\Lambda_{\mathbf{P}^1} \otimes \Lambda_{\mathbf{P}^1} = \Lambda_T$. Every entry of the resulting connection matrix lies in Λ_T, so applying χ entrywise commutes with the identification: the χ-specialized module of T is the tensor product of the modules specialized along the two restrictions of χ to the ruling classes f_1, f_2. Each factor is a projective line with specialized quantum relation $H^2 = \chi(Q^{f_i})$, and strict admissibility keeps that value nonzero, so by the argument of Proposition 5.9(ii) each factor has simple Euler spectrum and, by Lemma 5.20, trivial framed regular monodromy. As in the proof of Proposition 5.21 below, the Levelt–Turrittin formal decomposition is compatible with tensor products of differential modules, so the framed operator of T is the tensor product of two identities. The argument uses no relation between the two specialized values, which is what the product route buys: χ may identify the two ruling classes, and the discriminant of Lemma 5.12 then vanishes.

Let $a \geq 1$. Lemma 5.14 gives $\operatorname{disc} P_a \neq 0$, so by Lemma 5.13(a) every spectral block of Euler multiplication has rank one, and Lemma 5.20 makes the framed regular monodromy trivial on each. Hence no primitive sixth root occurs. □

Remark 5.16 (Where the tagging hypothesis is still used). Proposition 5.9 shows that Hypothesis 5.7T is not needed for nef-canonical centers, for $\mathbf{P}^1$ or $\mathbf{P}^2$, or—assuming only Hypothesis 5.7R—for ruled surfaces over positive-genus curves, and Proposition 5.15 removes it for the rational geometrically ruled centers as well. Those cases exhaust the minimal surfaces, and also cover F_1, which is geometrically ruled without being minimal. In the present proof of four-dimensional birational invariance, the substantive use of Hypothesis 5.7T is therefore confined to surface centers that are neither minimal nor geometrically ruled. Every such surface is an iterated point blowup, so eliminating it there would require a separate support or base-change statement for applying the blowup comparison after the external center specialization; no such statement is assumed here. The quartic of Lemma 5.12 was derived for the Hirzebruch surfaces and is not available for such a center.

Hypothesis 5.7T (Divisor-tagging specialization invariance)**.** *Let $\chi : \Lambda_T \to A$ be strictly Novikov-admissible. Choose integral divisors $D_1, \ldots, D_\rho$ whose pairings separate $N_1(T)$ and define*

$$\chi_{\boldsymbol{t}}(Q^d) = \chi(Q^d) \exp\left(\sum_i t_i (D_i \cdot d)\right).$$

After passing to algebraic closures of the relevant fraction fields, the intrinsic framed operator of the χ-specialized small connection and the tagged module have the same algebraic multiplicities at $e^{\pi i/3}$ and $e^{-\pi i/3}$.

This hypothesis concerns only the specialization step in this tagging family. It does not assert reconstruction invariance or invariance under an arbitrary divisor-valued bulk deformation.

Lemma 5.17 (Divisor tagging, conditional)**.** *Assume Hypothesis 5.7T. Let $\chi : \Lambda_T \to A$ be strictly Novikov-admissible. If the intrinsic small even quantum connection of T has no primitive-sixth framed monodromy, neither does its specialization along χ.*

Proof. If $N_1(T) = 0$, then the numerical effective monoid is $\{0\}$, so the coefficient map is already injective and there is nothing to tag. Assume otherwise. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(T)$, and introduce formal variables $\boldsymbol{t} = (t_1, \dots, t_\rho)$. Define

$$\chi_{\boldsymbol{t}}(Q^d) = \chi(Q^d) \exp\left(\sum_i t_i (D_i \cdot d) \right). \tag{5.6}$$

This formula extends continuously from monomials to the completed Novikov ring. Indeed, strict Novikov admissibility gives $v(\chi(Q^d)) = \ell(d)$, with ℓ positive and proper on the effective numerical monoid. Hence only finitely many classes contribute below any valuation cutoff, so the tagged sum converges coefficientwise in the complete ring $A[[\boldsymbol{t}]]$ and defines a continuous unital homomorphism. We now prove that it is injective. Take $f = \sum_d c_d Q^d \neq 0$ and put

$$\mu = \min\{v(\chi(Q^d)) : c_d \neq 0\} = \min\{\ell(d) : c_d \neq 0\}, \qquad S_\mu = \{d : c_d \neq 0,\ \ell(d) = \mu\}.$$

Properness in (5.5) makes S_μ finite and nonempty. Write in_v for the initial class in the associated graded. The degree-μ initial form is

$$\mathrm{in}_\mu(\chi_{\boldsymbol{t}}(f)) = \sum_{d \in S_\mu} c_d \,\mathrm{in}_v\Big(\chi(Q^d)\Big) \exp\left(\sum_i t_i (D_i \cdot d) \right) \ \in\ \mathrm{gr}_v(A)[[\boldsymbol{t}]],$$

and every displayed summand coefficient is nonzero. Every term outside S_μ has valuation strictly greater than μ, so a nonzero initial form proves $\chi_{\boldsymbol{t}}(f) \neq 0$. This is a finite combination of distinct exponential characters. The finitely many exponent vectors are distinct integral vectors. Choose $a = (a_i) \in \mathbf{Z}^\rho$ outside the finitely many integral hyperplanes on which two dot products agree, and substitute $t_i = a_i s$. The resulting exponents are distinct integers. Their derivatives at $s = 0$ form a Vandermonde matrix with nonzero integer determinant, which stays invertible over $\mathrm{Frac}(\mathrm{gr}_v A)$ because that field has characteristic zero. The characters are therefore linearly independent, and the initial form displayed above is nonzero.

The domain hypothesis on $\mathrm{gr}_v(A)$ is used twice, and only twice. It provides the field $\mathrm{Frac}(\mathrm{gr}_v A)$ of the Vandermonde step just made, and it makes v multiplicative on A, since initial forms of nonzero elements then have nonzero product. The valuation of a monomial image is $\ell(d)$ by hypothesis (5.5); multiplicativity is what preserves that value after multiplication by the tag, a unit of valuation zero. Terms with $\ell(d) > \mu$ therefore contribute nothing in degree μ, and each $d \in S_\mu$ contributes its initial form with c_d intact. Were the associated graded to have zero divisors, v would be a filtration rather than a valuation, and the displayed degree-μ identity could fail.

Injectivity of (5.6) embeds the intrinsic numerical Novikov fraction field into the fraction field of $A[[\boldsymbol{t}]]$, fixing $\mathbf{C}$. Therefore the module obtained by the tagged map is a scalar extension of the intrinsic small module, and their framed characteristic polynomials have the same primitive-sixth multiplicities after passage to a common algebraic closure.

On the other hand, setting $\boldsymbol{t} = 0$ in the tagged family gives the χ-specialized small connection. Hypothesis 5.7T is precisely the remaining assertion that this specialization does not change the two primitive-sixth multiplicities. Combining these two statements proves the lemma. Notice that no pro-Laurent or Hahn-field framed operator is invoked: the hypothesis is attached directly to the specialization step for which it is needed. □

Proposition 5.18 (Low-dimensional vanishing, conditional)**.** *Assume Hypothesis 5.7R, and Hypothesis 5.7T only for surface centers that are neither minimal nor geometrically ruled. Let T be a point, a smooth projective curve, or a smooth projective surface, and let χ be a strictly Novikov-admissible specialization, graded-monomial when $T \cong F_1$. Then*

$$\nu_6(T;\chi) = 0. \tag{5.7}$$

Proof. The direct specialized cases are already covered by Proposition 5.9: this includes points and curves of positive genus through the nef-canonical case, the two projective spaces $\mathbf{P}^1, \mathbf{P}^2$, every minimal surface with nef canonical bundle, and, assuming Hypothesis 5.7R, ruled surfaces over positive-genus curves.

The rational geometrically ruled surfaces, that is the Hirzebruch surfaces F_a, are covered by Proposition 5.15, again without Hypothesis 5.7T; the graded-monomial condition on χ is used only there, and only for F_1. Together these cases cover every minimal surface: a minimal smooth projective surface has nef canonical class, or is $\mathbf{P}^2$, or is geometrically ruled over a curve, by the Enriques–Kodaira classification. They also cover F_1, which is geometrically ruled but not minimal.

It remains to treat the surfaces that are neither minimal nor geometrically ruled. Work intrinsically first, before the external specialization. Every minimal surface has vanishing intrinsic invariant: apply the direct cases just listed to the identity map of Λ_T, which is strictly Novikov-admissible with $\ell = H \cdot (-)$ for an ample H, and graded-monomial because its associated graded is the monoid algebra of the effective monoid, whose monomials are a basis. A surface that is not minimal is obtained from a minimal one by point blowups, and under Hypothesis 5.7R the blowup formula (5.3) adds only point terms, which are zero; so its intrinsic small invariant vanishes as well. For this last case we use Hypothesis 5.7T through Lemma 5.17, which carries the intrinsic vanishing to χ, proving (5.7). □

Theorem 5.19 (Conditional birational invariance through dimension four)**.** *Assume Hypothesis 5.7R, and Hypothesis 5.7T only for surface centers that are neither minimal nor geometrically ruled. If Y and Y' are birational smooth projective complex varieties of the same dimension at most four, then*

$$\nu_6(Y) = \nu_6(Y'). \tag{5.8}$$

Consequently, for smooth projective threefolds,

$$Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1 \quad \Longrightarrow \quad \nu_6(Y) = \nu_6(Y').$$

Proof. Weak factorization decomposes a birational map between smooth varieties into blowups and blowdowns with smooth centers of codimension at least two [1]. In dimension at most four every nontrivial center has dimension at most two. The calculation following Definition 5.8 shows that every center specialization in (5.3) is strictly admissible, and Lemma 5.11 shows that it is graded-monomial. Proposition 5.18 therefore makes every center term zero, so the endpoint values agree at each step.

If Y, Y' are smooth threefolds and $Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1$, apply the first assertion in dimension four and (5.2):

$$2\nu_6(Y) = \nu_6(Y \times \mathbf{P}^1) = \nu_6(Y' \times \mathbf{P}^1) = 2\nu_6(Y').$$

Cancelling two in $\mathbf{Z}$ proves the stabilization assertion. $\square$

Lemma 5.20 (Multiplicity-one Euler blocks)**.** *Let T be smooth projective and consider its even quantum connection at an even bulk point over an algebraically closed coefficient field. If a generalized eigenspace of Euler multiplication has rank one, that block contributes only the eigenvalue* 1 *to framed regular monodromy. In particular, at the small point it contributes nothing to $\nu_6(T)$.*

Proof. Write the horizontal system at the chosen even bulk point in the standard form

$$z^2\partial_z Y = (U - z\mu)Y,$$

where U is Euler quantum multiplication and μ is the grading operator. The Frobenius property makes U self-adjoint for the Poincaré pairing, while μ is anti-self-adjoint. Generalized eigenspaces belonging to distinct eigenvalues of the self-adjoint operator U are orthogonal, so the pairing restricts nondegenerately to the one-dimensional block. If P is its spectral projector, anti-self-adjointness on this nondegenerate line therefore gives

$$P\mu P = 0.$$

Formally split this line from the other Euler eigenspaces by the normalized gauge $g = I + zg_1 + z^2 g_2 + \cdots$ whose positive coefficients are block-off- diagonal. The coefficient of z on the resulting scalar block is

$$-P\mu P + P[U, g_1]P = 0;$$

the logarithmic-derivative term of the gauge begins in order z^2. Consequently the scalar equation has the form

$$z^2\partial_z y = (u + z^2 h(z))y, \qquad h(z) \in K[[z]],$$

for the corresponding Euler eigenvalue u. Its normalized formal solution is $e^{-u/z}v(z)$ with $v(z) \in 1 + zK[[z]]$. No fractional power of the original loop coordinate and no logarithm occurs, so the framed regular monodromy on this block is the identity. $\square$

Proposition 5.21 (Products with projective space)**.** *For every smooth projective variety T and every $m \geq 0$,*

$$\nu_6(T \times \mathbf{P}^m) = (m+1)\nu_6(T).$$

This statement is unconditional.

Proof. The product formula for Gromov–Witten theory [5], followed by the numerical Novikov base change of Lemma 5.4, identifies the small numerical quantum connection of $T \times \mathbf{P}^m$ with the tensor-product quantum connection of the two factors. On even cohomology this is simply $H^{\mathrm{ev}}(T) \otimes H^{\mathrm{ev}}(\mathbf{P}^m)$, since projective space has no odd cohomology. After adjoining $q^{1/(m+1)}$ in the projective-space Novikov coefficient field, quantum multiplication by $c_1(\mathbf{P}^m) = (m+1)H$ has the $m+1$ distinct eigenvalues

$$(m+1)\zeta q^{1/(m+1)}, \qquad \zeta^{m+1} = 1.$$

By Lemma 5.20, each corresponding rank-one formal factor has trivial framed regular monodromy. The Levelt–Turrittin formal decomposition is compatible with tensor products of differential modules. Hence tensoring a formal factor of the quantum differential module of T with one of these rank-one projective-space factors only adds its scalar irregular exponential factor and leaves the original-z-disc regular framed operator of the T factor unchanged. The extension adjoining $q^{1/(m+1)}$ is in the coefficient field and is fixed by the z-turn. Consequently the product module has $m+1$ copies of every framed cyclotomic factor of T, counted with algebraic multiplicity. This gives the formula. $\square$

Corollary 5.22. *The small even framed invariant of projective three-space satisfies*

$$\nu_6(\mathbf{P}^3) = 0.$$

Proof. Apply Proposition 5.21 with $T = \mathrm{pt}$ and $m = 3$. $\square$

Proposition 5.23 (The cubic packet)**.** *For every smooth cubic threefold X,*

$$\nu_6(X) = 2. \tag{5.9}$$

This proposition is unconditional and uses neither Hypothesis 5.7R nor Hypothesis 5.7T.

Proof. Use the notation of Proposition 4.3 and put $F_X = \overline{\operatorname{Frac}\mathbf{C}[[q]]}$. That proposition separates the small even connection into two scalar blocks and the rank-two zero block (4.1), using only an algebraic Novikov extension and integral powers of the original loop coordinate. We treat the rank-two block first and return to the scalar blocks below.

For the rank-two block, seek a formal solution in the form

$$\widetilde{S}_3 = z^{\rho}(1+O(z)), \qquad \widetilde{S}_4 = cz^{\rho+1}(1+O(z)). \tag{5.9f}$$

The coefficient of $z^{\rho+1}$ in the first coordinate of (4.1), and that of $z^{\rho+2}$ in the second, give respectively

$$\rho = 2c - \frac{19}{18}, \qquad c(\rho+1) = \frac{19}{18}c - \frac{8}{81}. \tag{5.9g}$$

Eliminating c gives

$$\frac{1}{2}\left(\rho + \frac{19}{18}\right)\left(\rho - \frac{1}{18}\right) = -\frac{8}{81}, \qquad \text{hence} \qquad \rho^2 + \rho + \frac{5}{36} = 0. \tag{5.9h}$$

This is the coefficient comparison in Cai's displayed rank-two ansatz [7, p. 6].

The quadratic formula applied to (5.9h) gives

$$\rho = -\frac{1}{6},\ -\frac{5}{6}. \tag{5.9i}$$

These are exactly the two eigenvalues of the canonical modified residue R_X computed in Section 4. In particular, Proposition 4.11 identifies their squared difference with $\delta^\sharp(\alpha_X) = 4/9$; the present calculation independently recovers the same representatives from formal solutions. The constant matrix C adjoins only the algebraic Novikov coefficient r, and (4.6) and its inverse use only integral powers of z. Hence they preserve each exponent class in $\mathbf{Q}/\mathbf{Z}$: multiplication sends $z^\rho F_X[[z]]$ to itself, and the same holds for the inverse. Thus the two classes in (5.9i) remain the classes in the original-z-disc frame. By (5.0), the repaired framed convention assigns to a rational exponent ρ the eigenvalue $\operatorname{Exp}_V(\rho) = e^{2\pi i\rho}$. Therefore these two classes give

$$\operatorname{Exp}_V\left(-\frac{1}{6}\right) = e^{-\pi i/3}, \qquad \operatorname{Exp}_V\left(-\frac{5}{6}\right) = e^{-5\pi i/3} = e^{\pi i/3}. \tag{5.9j}$$

This agrees with Cai's rank-two calculation [7, p. 6].

Put $R = \mathbf{C}[r, r^{-1}]$. For completeness the scalar blocks may equivalently be written as

$$z^2\partial_z \widetilde{S}_\pm = (\pm 6r + h_\pm(z))\widetilde{S}_\pm, \qquad h_\pm(z) \in z^2 R[[z]]. \tag{5.9k}$$

Their normalized formal solutions are

$$\begin{aligned} \widetilde{S}_+ &= e^{-6r/z}u_+(z), \quad u_+(z) \in 1 + zR[[z]], \\ \widetilde{S}_- &= e^{6r/z}u_-(z), \quad\ \ u_-(z) \in 1 + zR[[z]]. \end{aligned} \tag{5.9l}$$

Indeed, after the displayed exponential is removed, $\partial_z u_\pm = z^{-2}h_\pm(z)u_\pm$. The coefficient $z^{-2}h_\pm(z)$ lies in $R[[z]]$, so integration with initial value one gives the asserted units. The exponentials and these units use only integral powers of the original z; neither adjoins a root of z nor produces a deck factor. Thus each block is unramified and has framed regular-monodromy eigenvalue 1, so neither contributes a primitive sixth root. This agrees with Cai's scalar solutions [7, pp. 5–6].

It remains to count the rank-two solutions. Write its ansatz as $\widetilde{S}_3 = z^\rho \sum_{n\geq 0} a_n z^n$ and $\widetilde{S}_4 = z^{\rho+1}\sum_{n\geq 0} b_n z^n$, with $a_0 = 1$. At coefficient n, the two equations have the form

$$L_{\rho+n}\begin{pmatrix} a_n \\ b_n \end{pmatrix} = \begin{pmatrix} \alpha_n \\ \beta_n \end{pmatrix}, \qquad L_s = \begin{pmatrix} s + 19/18 & -2 \\ 8/81 & s - 1/18 \end{pmatrix}, \qquad \det L_s = \left(s + \frac{1}{6}\right)\left(s + \frac{5}{6}\right). \tag{5.9m}$$

Here (α_n, β_n) depends only on coefficients of lower index; for $n = 0$ it is zero, and (5.9h) is exactly the condition that L_ρ have a kernel. The roots in (5.9i) are distinct modulo $\mathbf{Z}$, so $L_{\rho+n}$ is invertible for every $n \geq 1$. For either root, $\ker L_\rho$ is one-dimensional and its first coordinate is nonzero; the normalization $a_0 = 1$ fixes b_0. The recursion therefore gives one formal solution for each root and no further solution in its exponent class. The two resulting solutions are independent, and exhaust the rank-two block.

Consequently each eigenvalue in (5.9j) has algebraic multiplicity one, while (5.9l) contributes neither primitive sixth root. Hence

$$\nu_6(X) = 2. \tag{5.9n}$$

This is a calculation in the small even connection, so no odd-cohomology block enters the count. Moreover, $N_1(X) = \mathbf{Z}\ell$, and the numerical Novikov ring is exactly $\mathbf{C}[[Q^\ell]] = \mathbf{C}[[q]]$; thus the numerical-Novikov convention does not alter the calculation.

The shared reduction in Proposition 4.3 is proved internally from Cai's displayed small-even matrix; the argument above uses only that reduction and the ordinary formal recursion in the original loop coordinate.

□

Corollary 5.24 (Formal-germ persistence of the cubic packet)**.** *After extending the cubic Novikov coefficient field so that $r = (3q)^{1/2}$ is a unit, the primitive-sixth multiplicity is equal to 2 throughout the formal even bulk germ at the small hyperplane point.*

Proof. At the closed point the Euler eigenvalues are $6r, -6r, 0$, with the zero cluster a single rank-two Jordan block and the two nonzero clusters of rank one. The differences from the zero eigenvalue are units, so Proposition 5.5 applies to the zero packet and freezes its two exponent classes $-1/6, -5/6$. The two simple packets remain rank one on the formal germ and contribute no primitive sixth root by Lemma 5.20. Proposition 5.23 sets the value at the closed point equal to 2. □

Corollary 5.25 (Framed count after one product stabilization)**.** *For every smooth cubic threefold X,*

$$\nu_6(X \times \mathbf{P}^1) = 4.$$

This statement is unconditional.

Proof. Combine Proposition 5.21 with Proposition 5.23. □

Proof of Theorem 1.7. Corollary 5.25 gives $\nu_6(X \times \mathbf{P}^1) = 4$ unconditionally. Let Y be a rational smooth projective fourfold. Then Y is birational to $\mathbf{P}^4$, so Theorem 5.19 gives $\nu_6(Y) = \nu_6(\mathbf{P}^4)$, while Proposition 5.21 with $T = \mathrm{pt}$ and $m = 4$ gives $\nu_6(\mathbf{P}^4) = 5\nu_6(\mathrm{pt}) = 0$. Since $4 \neq 0$, the fourfold $X \times \mathbf{P}^1$ is not rational. □

This is the framed-monodromy proof of one-step irrationality. It is independent of the atomic argument of Section 4, which proves the same conclusion without either hypothesis.

5.1 A genus-eight Fano consequence

Kuznetsov associates to every smooth prime Fano threefold V of genus eight a smooth Pfaffian cubic threefold X, together with rank-two vector bundles U on V and E^* on X, whose projectivizations are related by a flop [23, Theorems 2.17–2.18].

Corollary 5.26 (One stabilization of V_{14})**.** *For every smooth prime Fano threefold V of genus eight,*

$$V \times \mathbf{P}^1 \textit{ is irrational.}$$

Assuming Hypothesis 5.7R, and Hypothesis 5.7T only for surface centers that are neither minimal nor geometrically ruled, one moreover has

$$\nu_6(V) = 2. \tag{5.11}$$

Proof. The flop gives a birational map between $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$. A projective bundle is birational to the product of its base with the corresponding projective space, so $V \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$. The latter is irrational by Theorem 1.1, proving the unconditional assertion.

Assume now Hypothesis 5.7R, and Hypothesis 5.7T only for surface centers that are neither minimal nor geometrically ruled. Both projectivizations are smooth projective fourfolds. By Theorem 5.19, Proposition 5.7, and Proposition 5.23,

$$2\nu_6(V) = \nu_6(\mathbf{P}_V(U)) = \nu_6(\mathbf{P}_X(E^*)) = 2\nu_6(X) = 4.$$

Hence $\nu_6(V) = 2$. □

The two-hypothesis framed refinement is sharp at this level. In dimension five a weak-factorization center may have dimension three and may itself carry the cubic packet, so the low-dimensional vanishing used in Theorem 5.19 no longer prevents cancellation. The ordinary atom obstruction has the parallel limitation recorded after Theorem 1.1: from the second stabilization onward the cubic atom is already allowed as a lower-dimensional representative.

6 The separation theorem

The irrationality input in this section is now unconditional: Theorem 1.1 applies to every smooth cubic threefold. The universal CH_0-triviality inputs are independent classical or cycle-theoretic statements, and pass to the product by the projective-bundle formula for Chow groups.

Proof of Corollary 1.2. Voisin produces a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds along which the primitive minimal class $\Theta^4/4!$ is algebraic, hence along which CH_0 is universally trivial [30, Theorem 4.5 and Lemma 4.6]. The projective-bundle formula supplies universal CH_0-triviality of the product, and Theorem 1.1 supplies its irrationality. $\square$

Proof of Corollary 1.3. The Fermat equation is a sum of five forms in separated single variables, so [11] gives universal CH_0-triviality; the projective-bundle formula passes it to the product. Irrationality follows from Theorem 1.1. $\square$

Proof of Corollary 1.4. The cubics of [32, Theorem 3.3] carry unirational parametrizations of degrees two and three, and [32, Corollary 3.5] keeps both degrees on $X \times \mathbf{P}^m$. Parametrizations of coprime degrees give a decomposition of the diagonal and hence universal CH_0-triviality [32, Section 1]. Theorem 1.1 gives irrationality of $X \times \mathbf{P}^1$. $\square$

Proof of Theorem 1.5. Corollary 3.6 gives universal CH_0-triviality of X_b, and the projective-bundle formula gives the same property for $X_b \times \mathbf{P}^1$. Theorem 1.1 proves that this fourfold is irrational. Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the A_5-component: Hartlieb proves that the closure of its image is a one-dimensional special subvariety [17, Proposition 5.7]. Finally, Proposition 3.9 shows that all but finitely many points of this moduli curve lie outside the separated-variable locus. $\square$

The two mechanisms are now independent without any standing hypothesis. On the cycle side, algebraicity of $\Theta^4/4!$ removes the universal-CH_0 obstruction. On the quantum side, the ordinary cubic Hodge atom α_X, decorated by $\delta^\sharp(\alpha_X) = 4/9$, cannot occur in dimension at most two and therefore obstructs rationality of the stabilized fourfold. The special A_5-geometry enters only in the cycle argument; the atom obstruction applies to every smooth cubic threefold. The cycle construction has an independent payoff: by Proposition 3.9, for all but finitely many moduli points of the A_5-curve this universal CH_0-triviality is not accounted for by Colliot-Thélène's separated-variable criterion.

The framed invariant ν_6 supplies a finer, partially conditional view of the same small quantum packet. Its values $\nu_6(X) = 2$ and $\nu_6(X \times \mathbf{P}^1) = 4$ are unconditional for every smooth cubic threefold by Proposition 5.23 and Corollary 5.25. Hypothesis 5.7R is used only for the residual reconstruction-tail step in the operation formulas. The direct specialization arguments of Propositions 5.9 and 5.15 remove Hypothesis 5.7T for nef-canonical targets, $\mathbf{P}^1$, $\mathbf{P}^2$, every rational geometrically ruled surface (for F_1, on the specializations that arise as blowup centers), and, assuming 5.7R, ruled surfaces over positive-genus curves; 5.7T remains only for surface centers that are neither minimal nor geometrically ruled. Conditional on 5.7R and on 5.7T for those centers, ν_6 is birationally invariant through dimension four. Neither hypothesis is a condition on any separation statement in this section.

For comparison, the two threefold-level quantities are logically independent. If $U(Y)$ denotes the indicator of universal CH_0-triviality, Corollary 5.22 gives $\nu_6(\mathbf{P}^3) = 0$ unconditionally, while Proposition 5.23 gives $\nu_6 = 2$ for every cubic. Consequently

$$\big(U(Y), \nu_6(Y)\big) = \begin{cases} (1,0), & Y = \mathbf{P}^3, \\ (1,2), & Y \text{ is a smooth member of the } A_5\text{-family}, \\ (0,2), & Y \text{ is a very general cubic threefold.} \end{cases}$$

Voisin's criterion [30, Corollary 4.4] and the evenness theorem [12, Theorem 1.3 and Corollary 1.4] give the last line. Thus neither universal CH_0-triviality nor ν_6 determines the other.

For $X \times \mathbf{P}^m$ with $m \geq 2$, neither obstruction used here proves stable irrationality. The ordinary atom criterion permits a dimension-three representative, which the cubic atom already has, while the conditional framed weak-factorization argument may encounter dimension-three centers carrying primitive-sixth packets. Likewise, the fibrewise minimal-class argument gives no relative universal cycle over the pencil. These are separate problems and are not implicit in the theorem above.

Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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